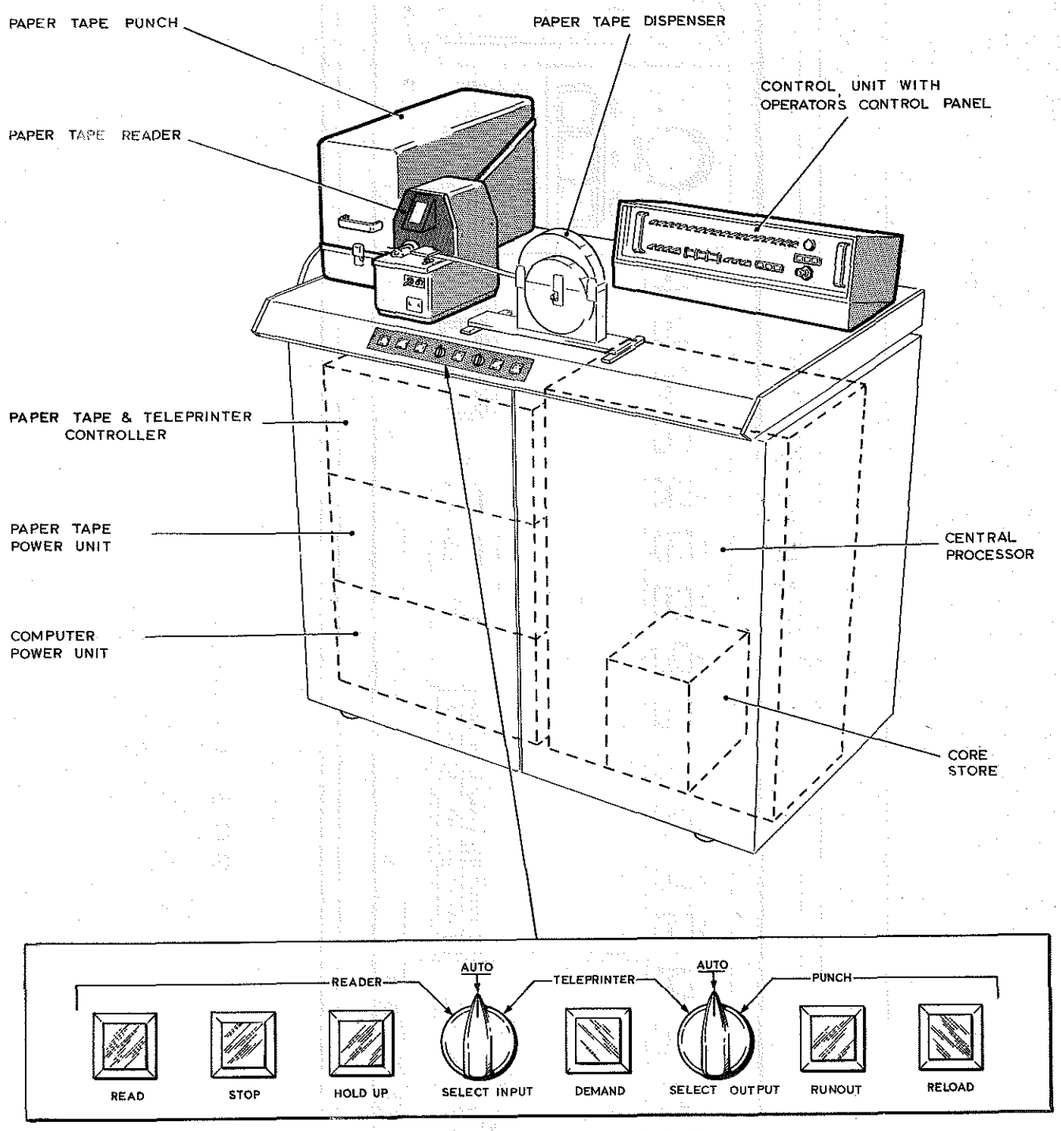


PAPER TAPE & TELEPRINTER CONTROL PANEL

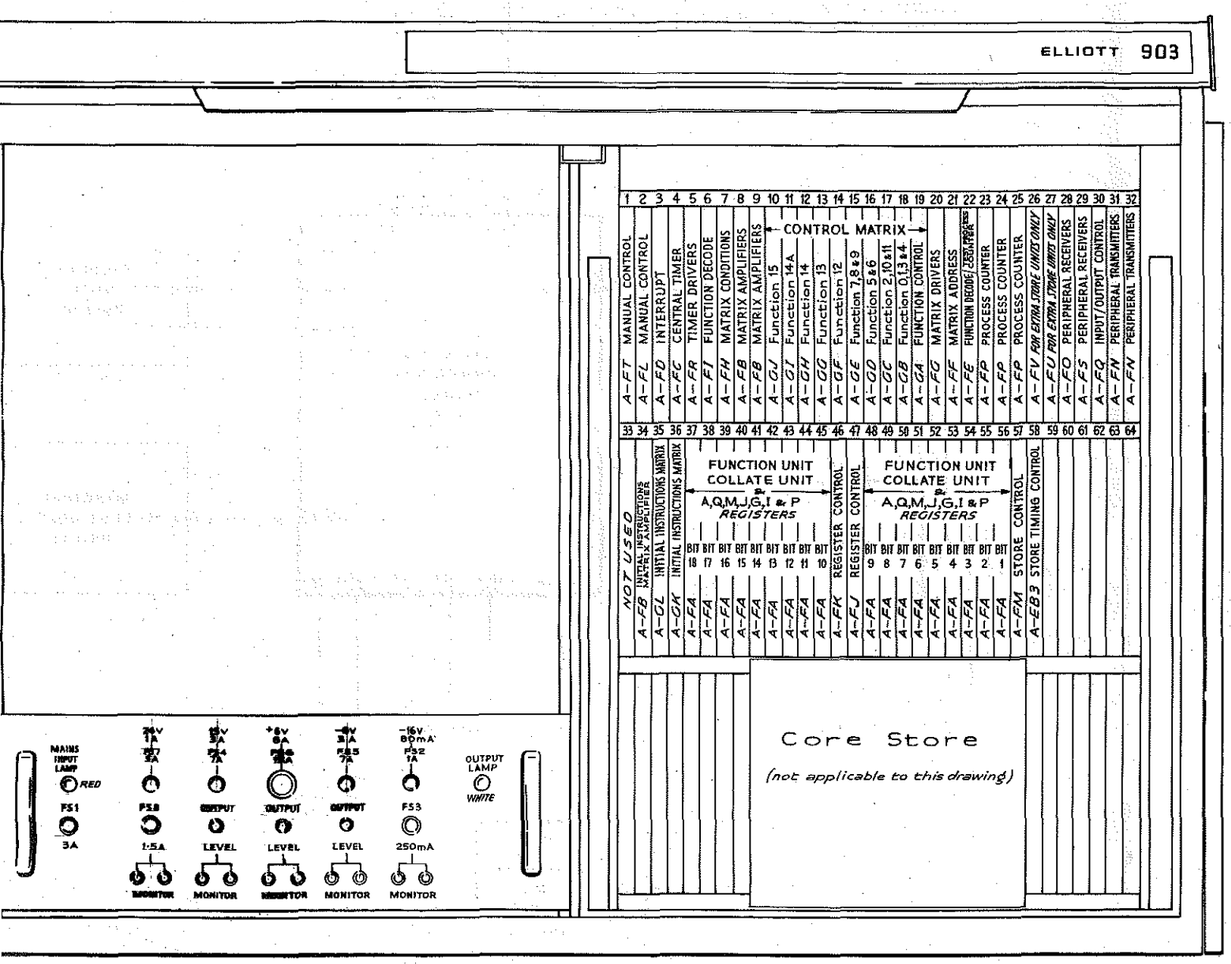
THIS IS THE DESK UNIT FROM THE FRONT AND SHOWS THE INTERNAL STORAGE OF EQUIPMENT AS WELL AS THE LAYOUT OF EQUIPMENT BUILT INTO DESK TOP

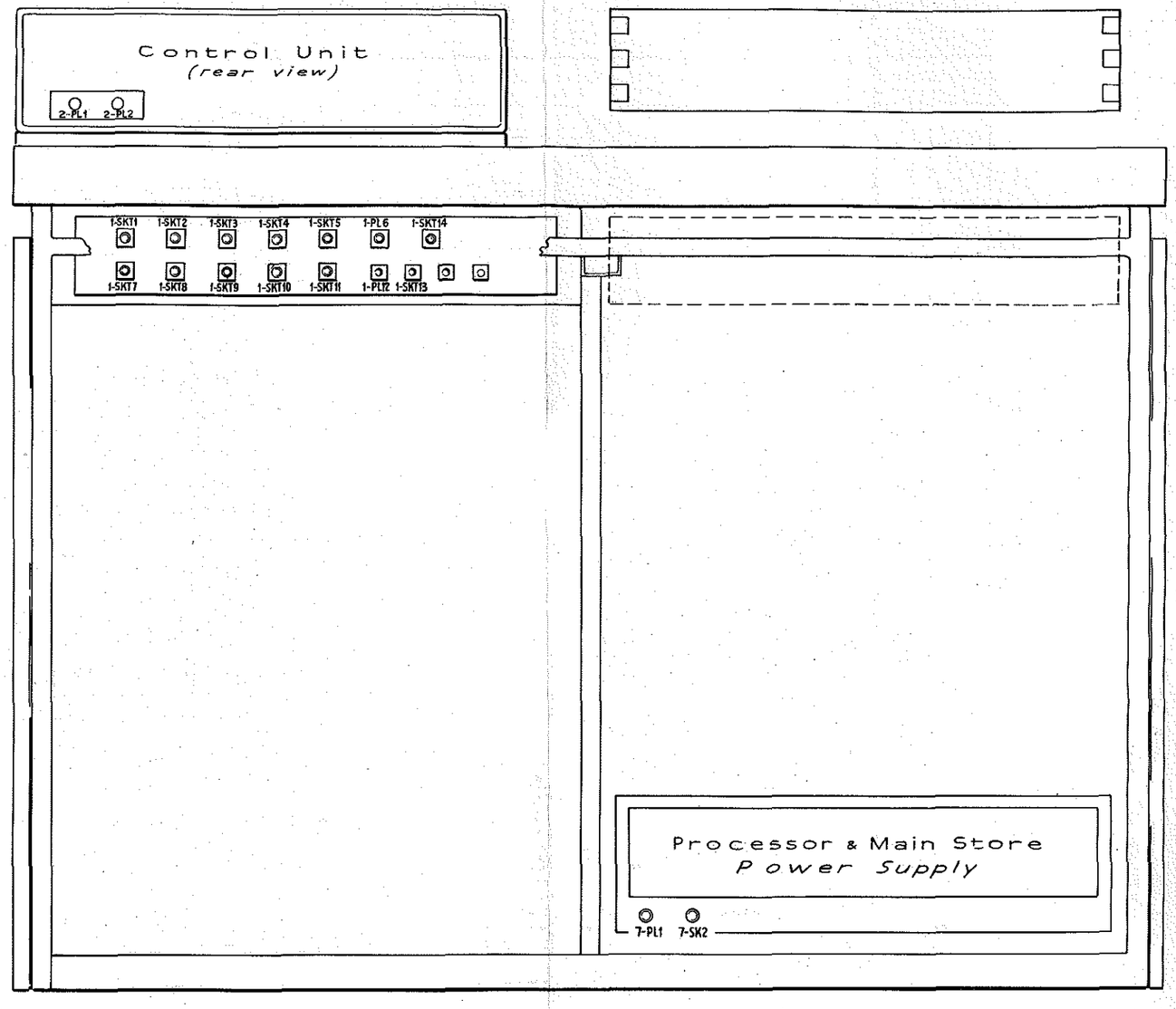
OVERALL VIEW

Figure 1 (ISSUE 2)

Figure 2

(Issue 2)





RAL PROCESSOR DISPOSITION

Figure 3 (ISSUE 2)

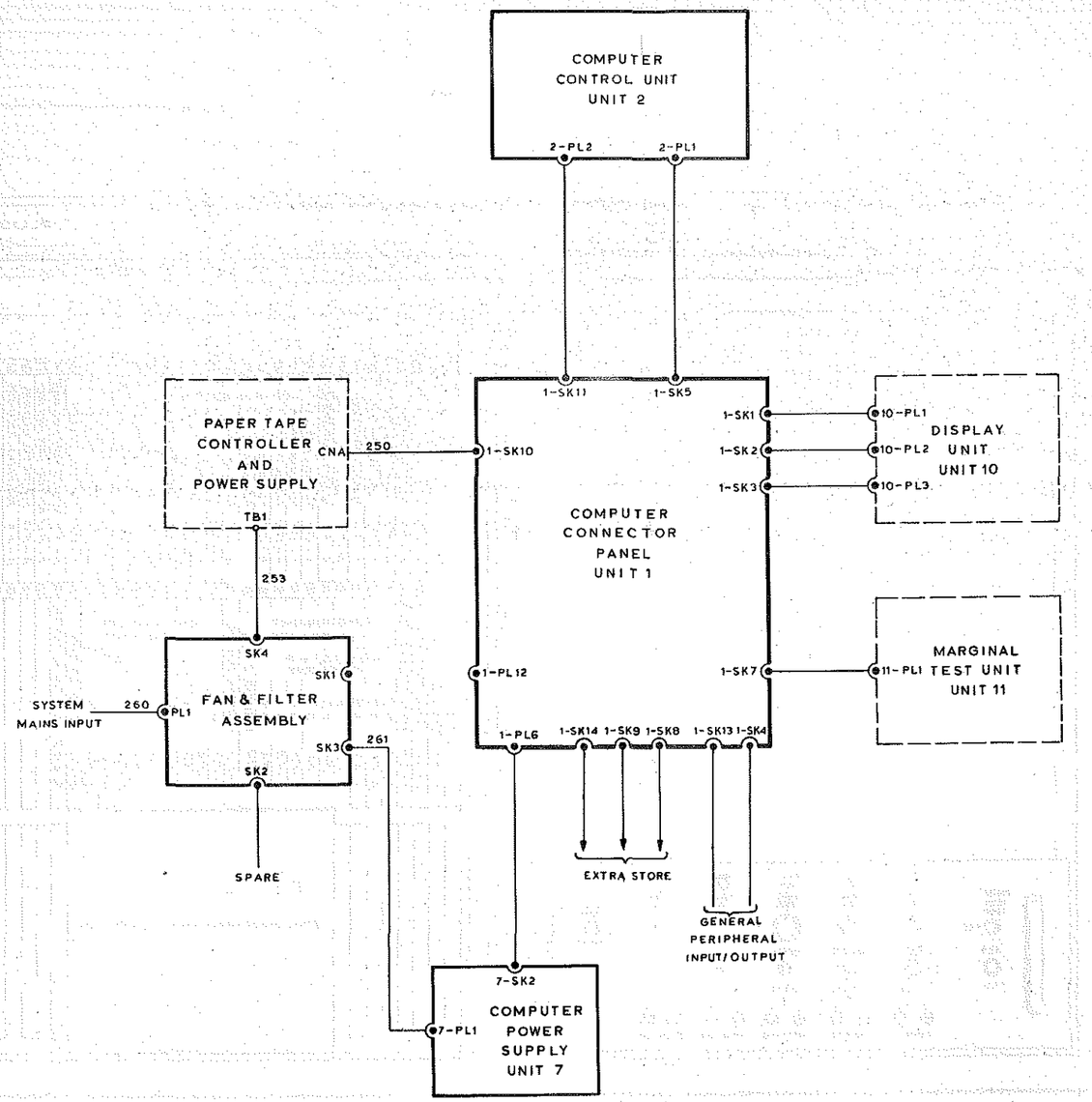


Figure 4 (ISSUE 2)

WORD GENERATOR

MODIFIER | FUNCTION BITS | ADDRESS BITS

SW35 SW34 SW33 SW32 SW31 SW30 SW29 SW28 SW27 SW26 SW25 SW24 SW23 SW22 SW21 SW20 SW19 SW18

WG-18k WG-16k WG-15k WG-14k WG-13K WG-12k WG-11k WG-10k WG-9k WG-8k WG-7k WG-6k WG-5k WG-4k WG-3k WG-2k WG-1k

All word generator outputs come out on 2—PL2.

SW5 OV — RESET k
SW7 — OFF
SW7 — IN
SW6 — ON
WG-17 k
WG-17k
JUMP k
JUMP k
STOP k
RESTART k
RESTART k
OWG k
OWG k
CINS k
CINS k
EWG k
EWG k
CS k
SMN k
OS k
TCS k
TEST k
AUTO k

COMPUTER MODE SWITCH SW1
c = AUTO
b = OPERATE
a = TEST

NOTE: PINS WHICH ARE PREFIXED TL are connections on the T-L printed circuit board mounted inside the Control Unit.

1(A-FT) 4(A-FC) 2(A-FL) 5(A-FR) 6(A-FT) 22(A-FE)

OWG EWG STOP +6V +2·4V WFR CON CINS k CIN

JUMP k JUMP 15 JUMP JUMP 16 JP JIA AUTO k AUTO RESTART k STOP k STOP RESET/2 OS k OS RESTART CS k CS OBEY OWG k OWG EWG k EWG TEST k ENTER W17 WTM a

R1 3·9K R2 3·9K STOP 1 STOP 2 OIS (ORDER STOP)

R4 2·2K R5 2·2K R6 100K C1 47μF R7 2·2K R8 2·2K R9 2·2K R10 2·2K

RESTART MST (MANUAL STEP TIMER) MARGINS +2·4V/1 VR

END OF RESET SAO (SET MATRIX ADDRESS 0)

STC ND SHO NOT READY RESET 1 RESET 2 RESET 1 RESET 2 R3 3·9K R4 3·9K R5 3·9K

SB RESET k RESET RESET JIA STC PSC RDY RDY

TEST (TO 4(A-FC) PIN 8) CIS (CYCLE STOP)

MANUAL CONTROLS — KEYS & LOGIC

Figure 5 (ISSUE 2)

900
4.1.2.

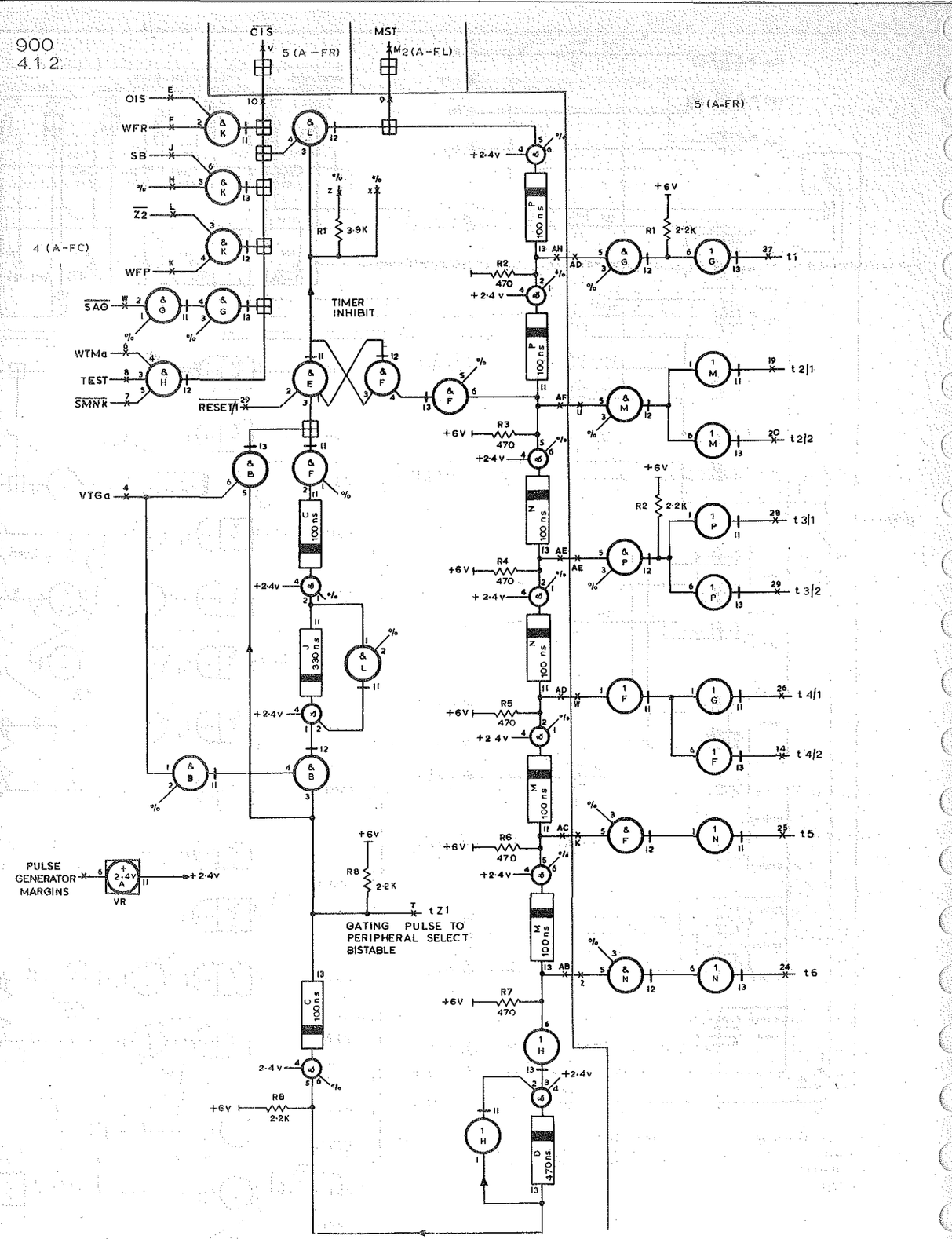
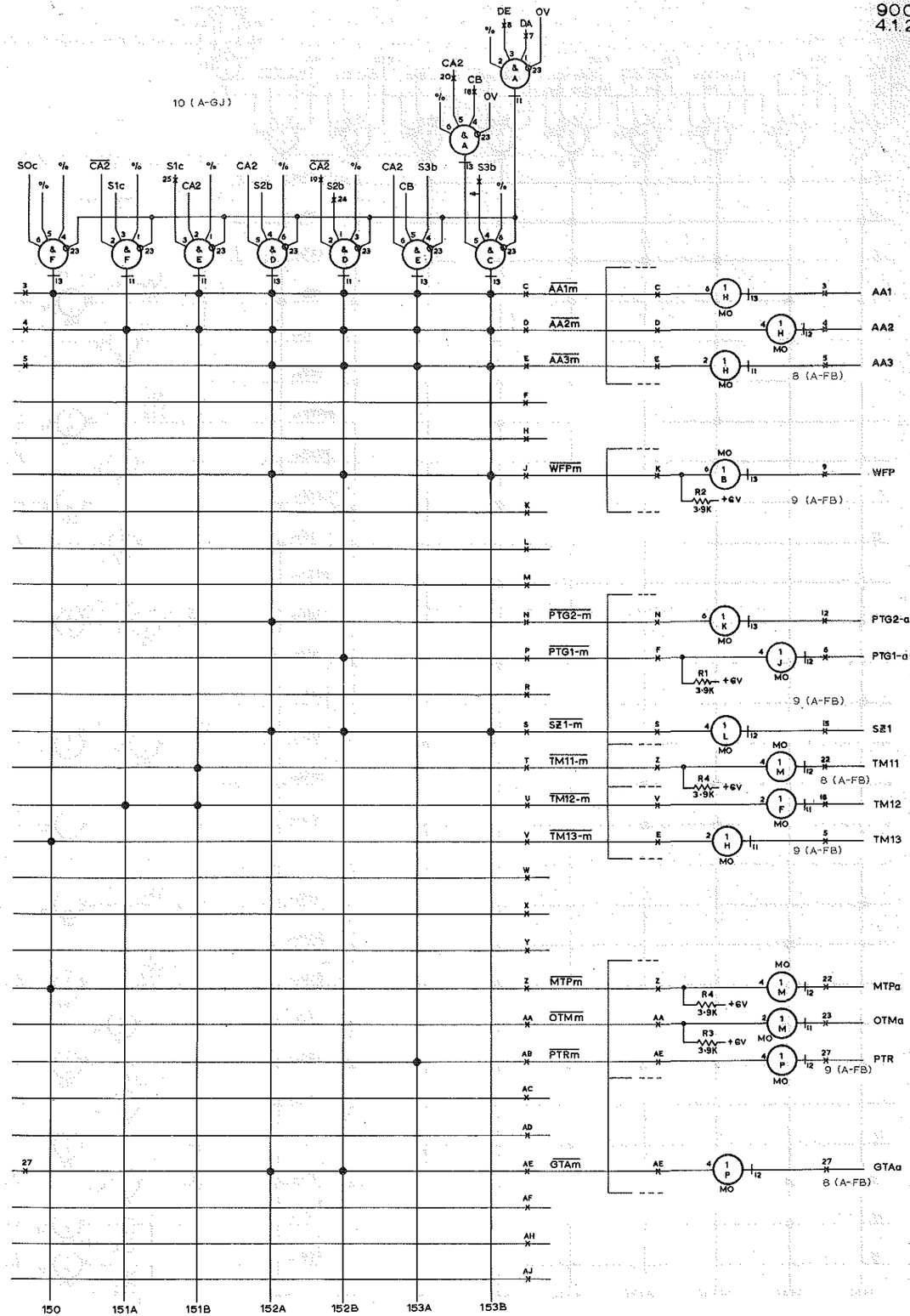


Figure 6 (ISSUE 2)

CENTRAL TIMER



CONTROL MATRIX FUNCTION 15

Figure 7 (ISSUE 2)

900
4.1.2.

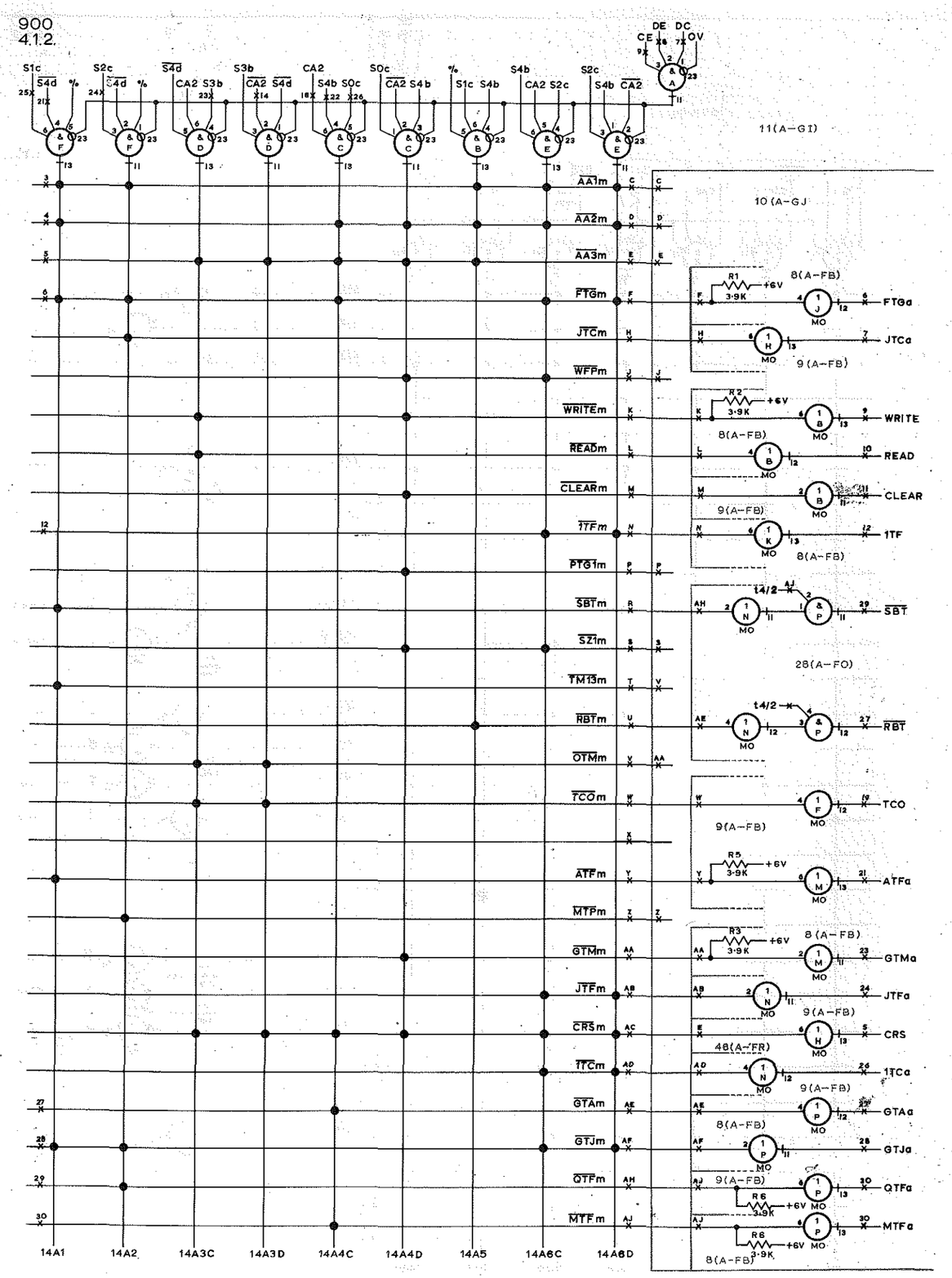


Figure 8 *(ISSUE 2)*

CONTROL MATRIX FUNCTION 14A

CONTROL MATRIX – FUNCTION 14

Figure 9 *(ISSUE 2)*

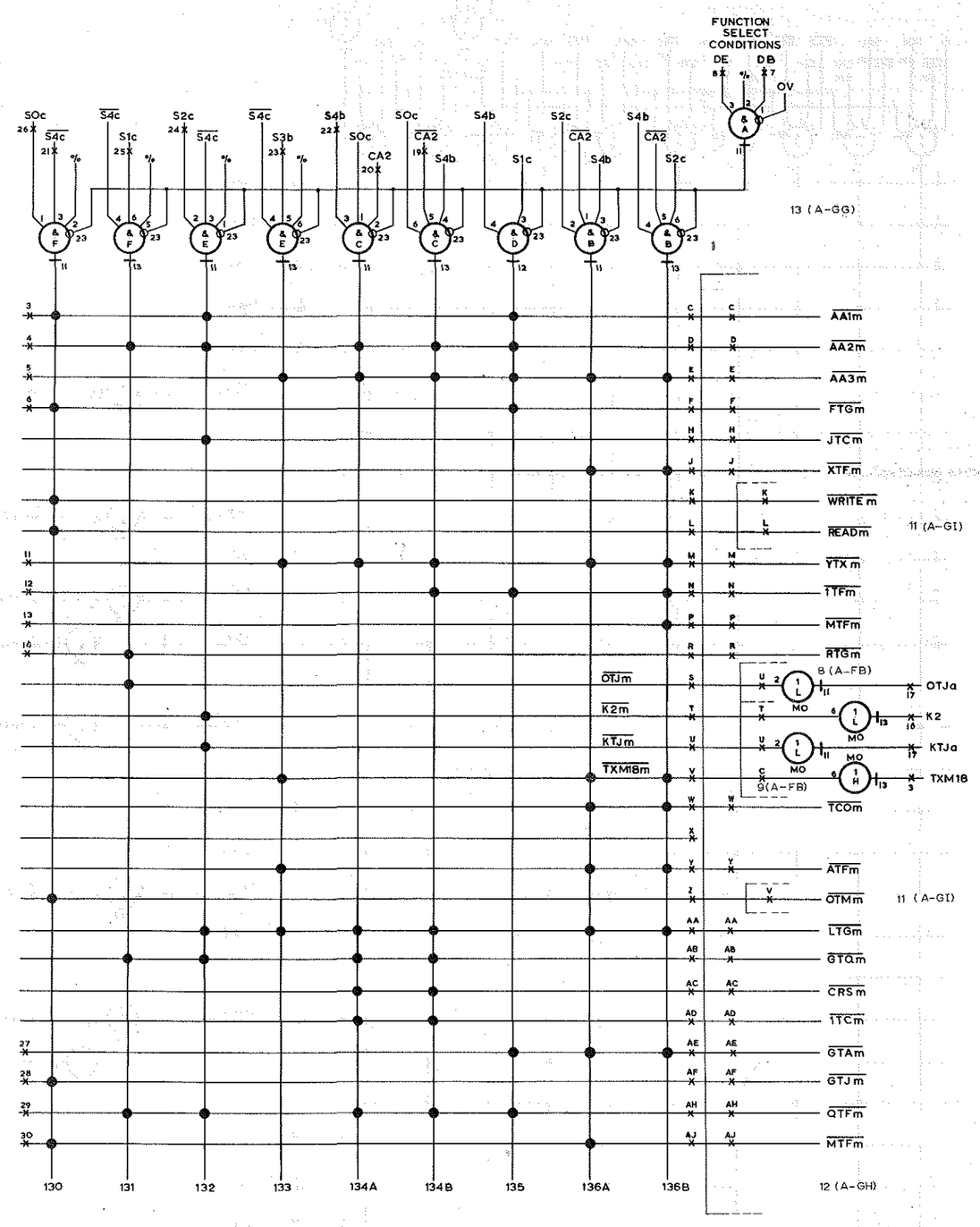
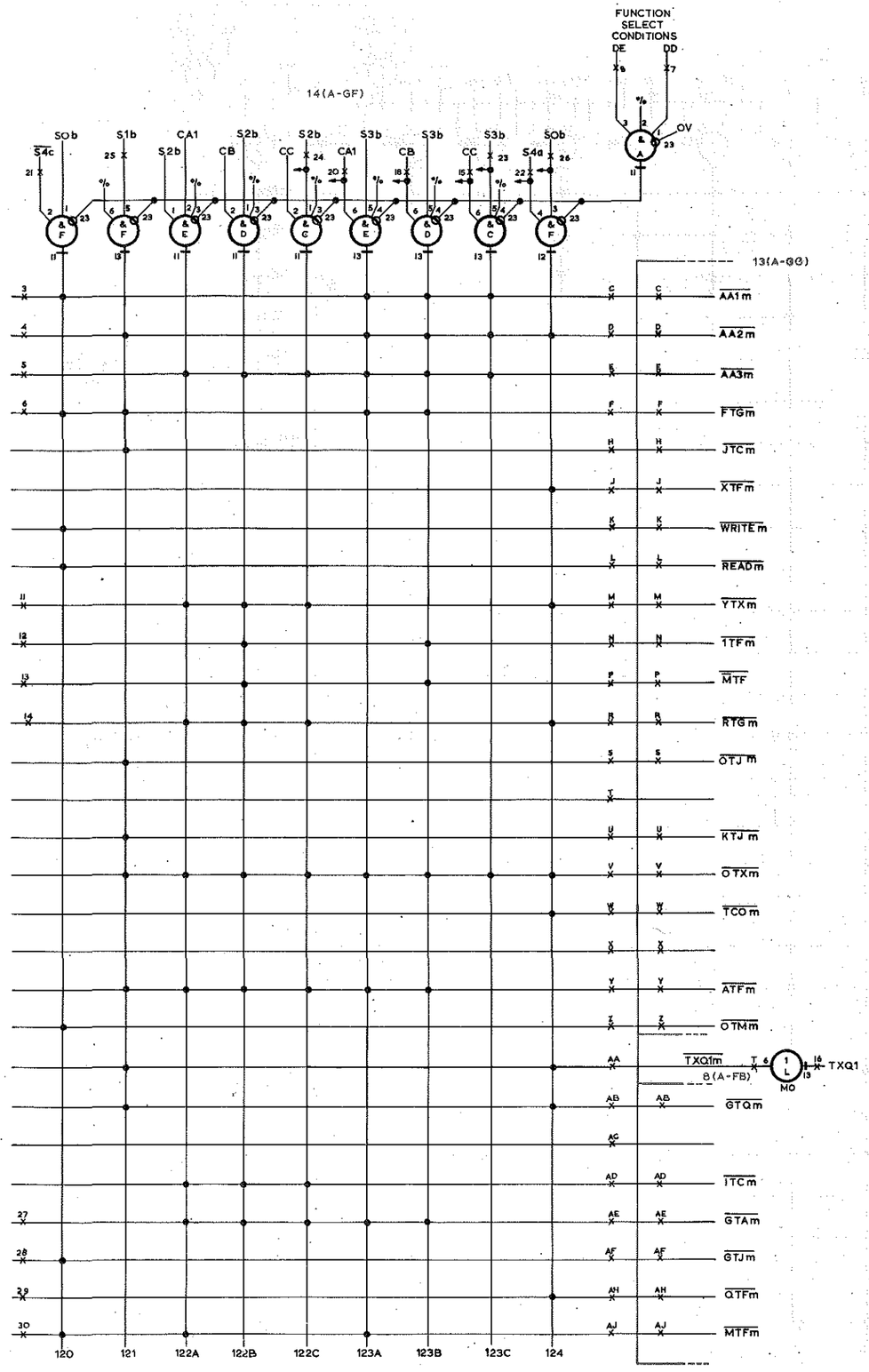


Figure 10 *(ISSUE 2)*

CONTROL MATRIX FUNCTION 13



CONTROL MATRIX FUNCTION 12

Figure 11 (ISSUE 2)

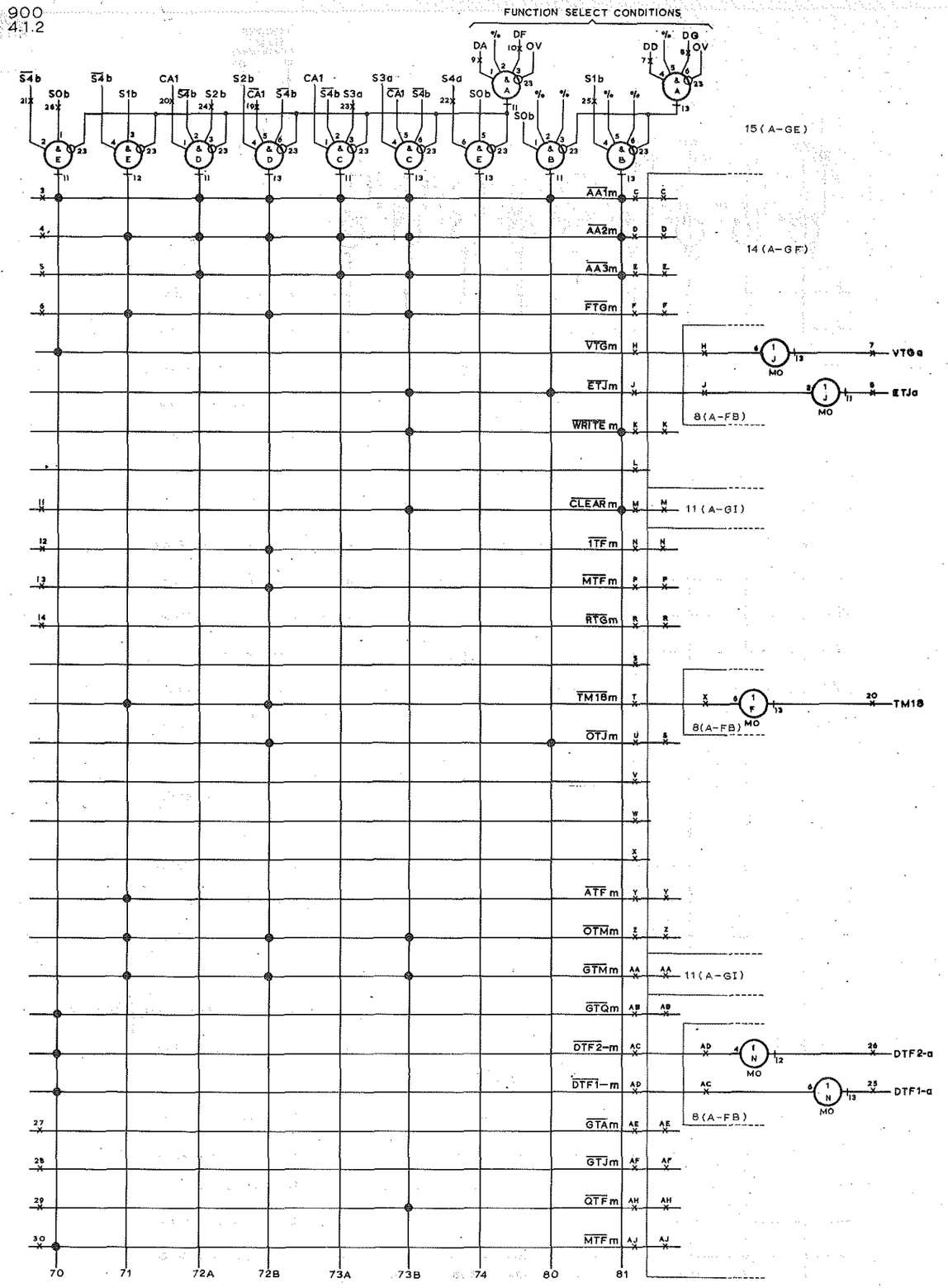
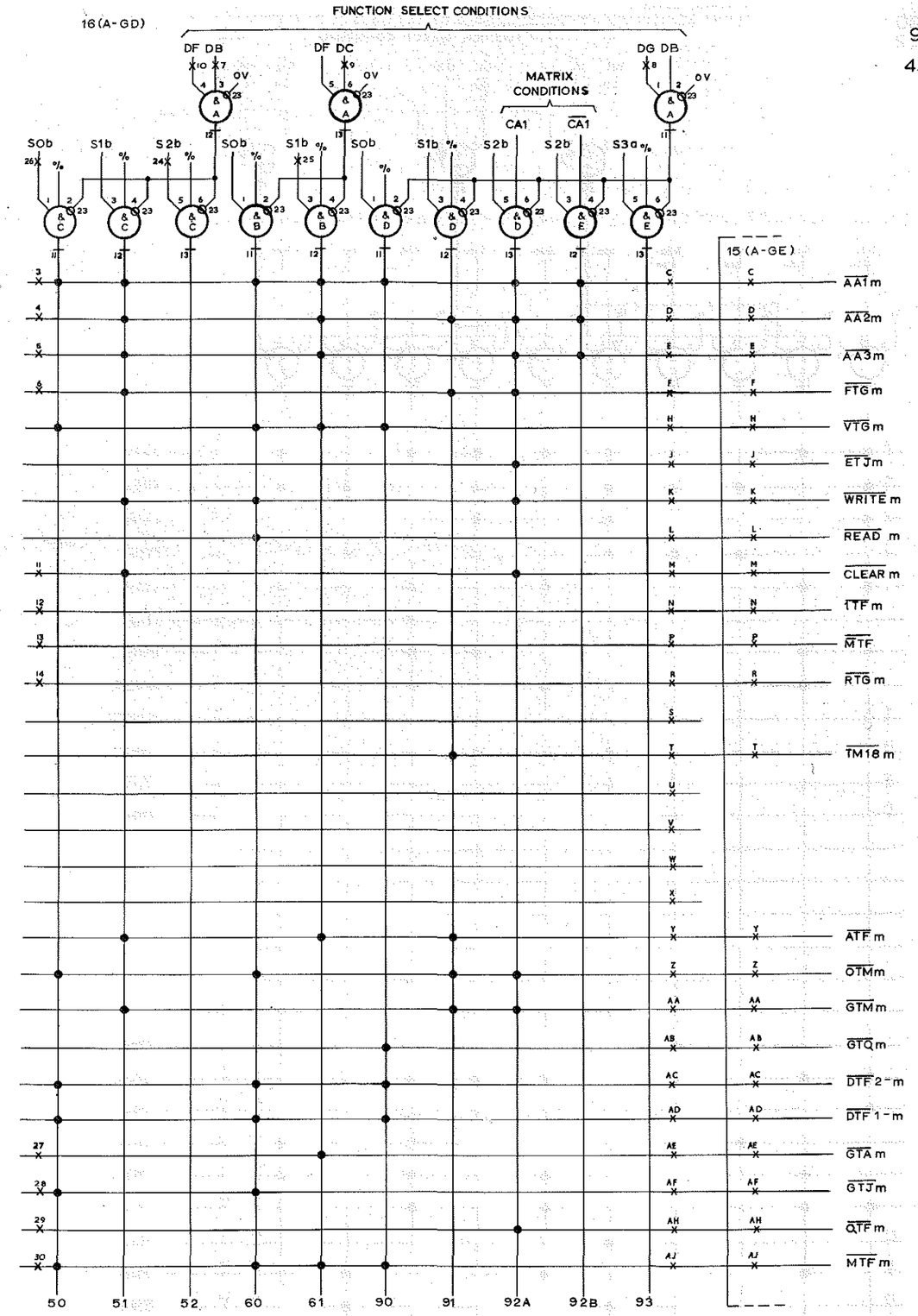


Figure 12 *(ISSUE 2)*

CONTROL MATRIX FUNCTIONS 7, 8.



CONTROL MATRIX FUNCTIONS 5,6 & 9

Figure 13 (ISSUE 2)

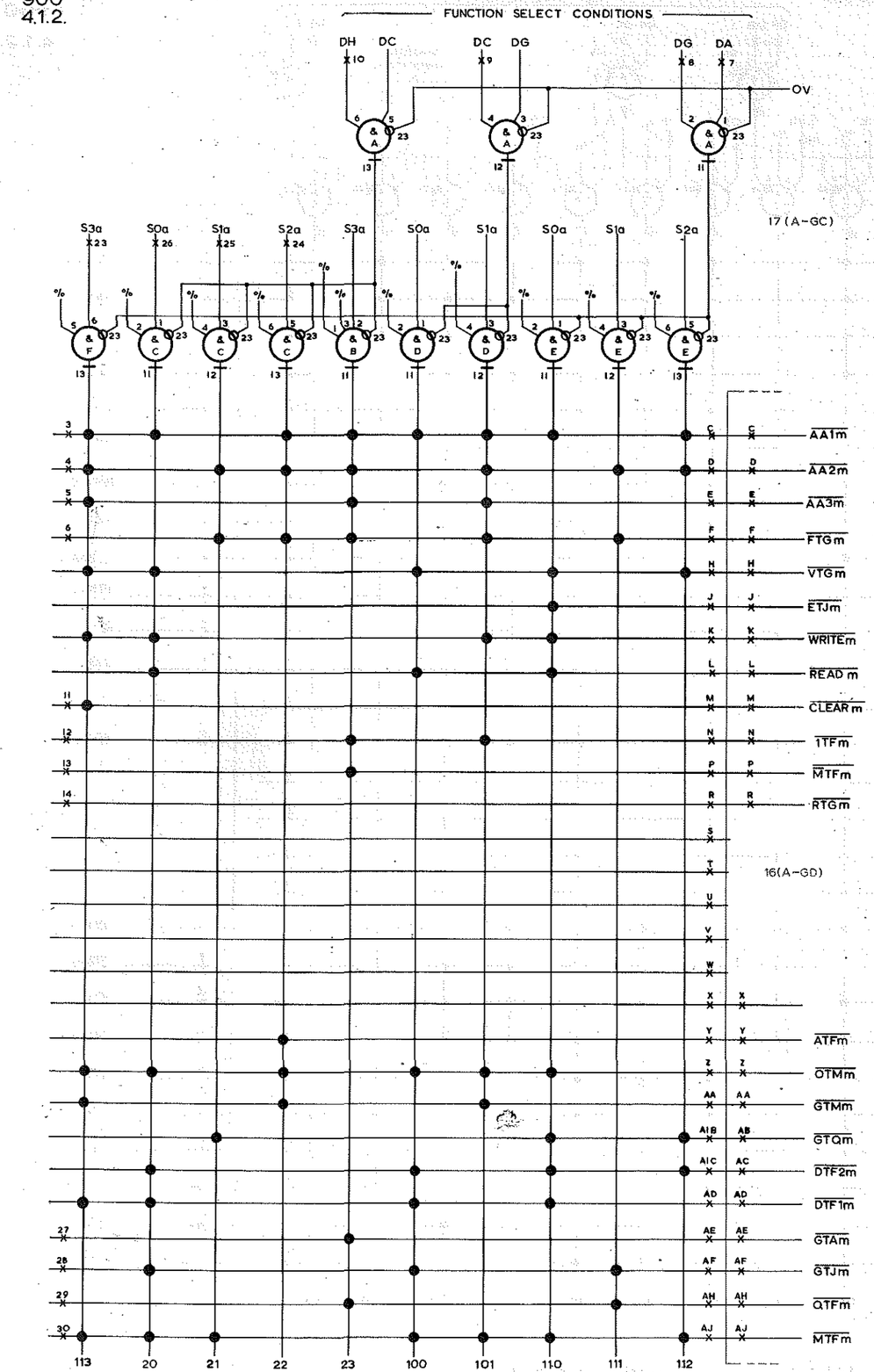
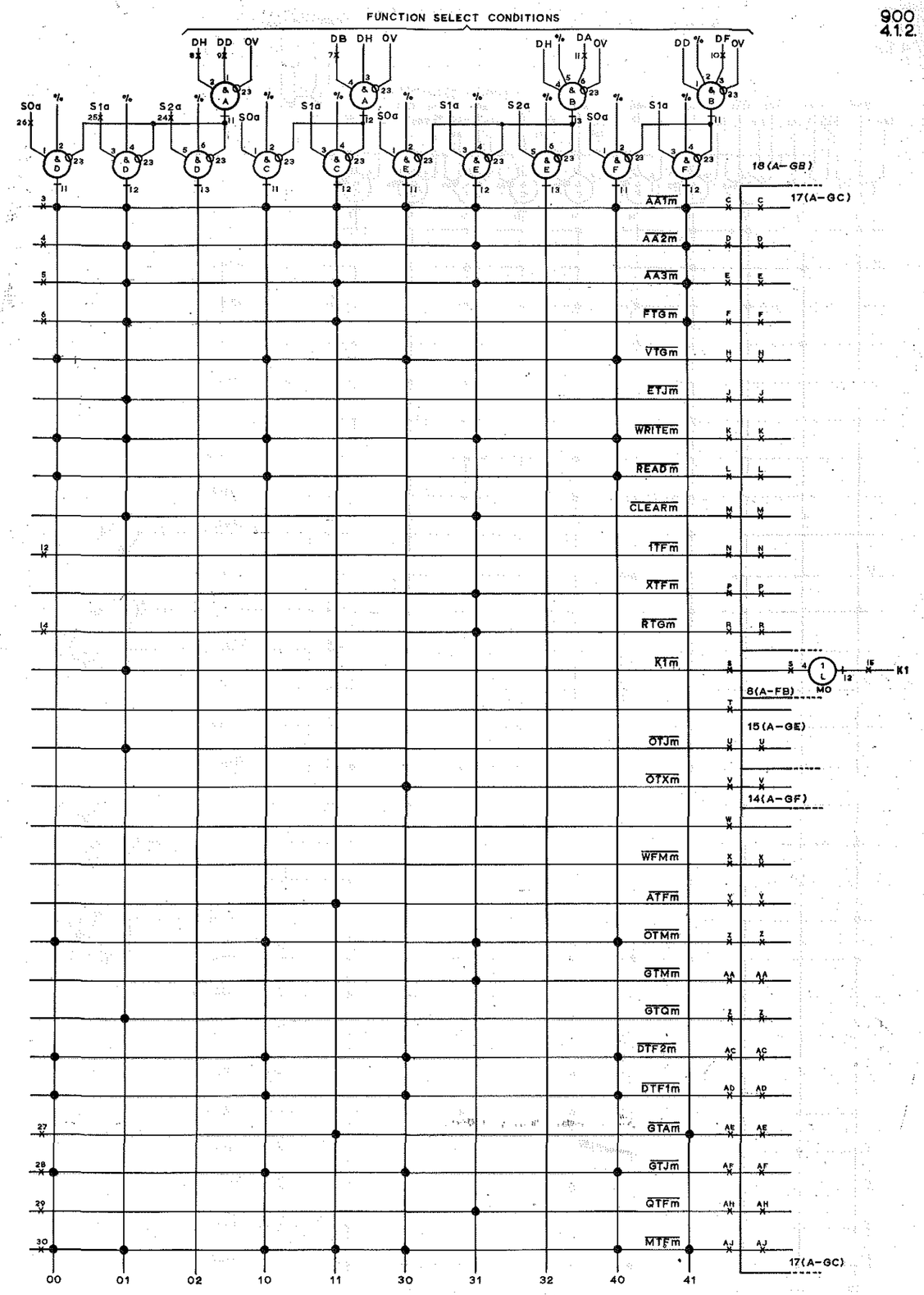


Figure 14 *(ISSUE 2)* CONTROL MATRIX FUNCTIONS 2, 10, 11.



CONTROL MATRIX FUNCTIONS 0,1,3,4.

Figure 15 *(ISSUE 2)*

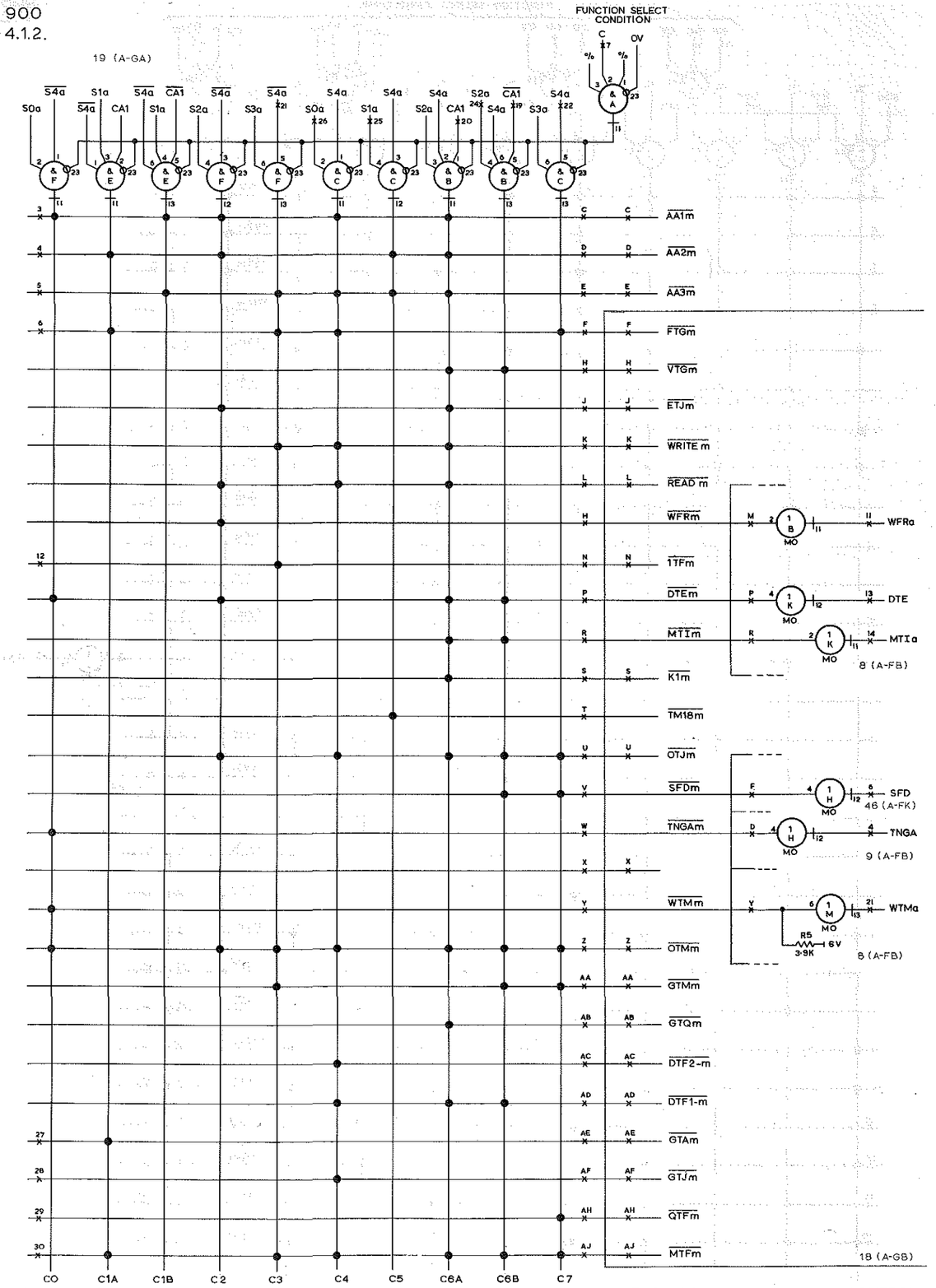


Figure 16 *(ISSUE 2)*

CONTROL MATRIX FUNCTION CONTROL

7(A-FH)

22(A-FE)

6(A-FI)

20(A-FG)

6(AF-I)

22(A-FE)

21(A-FF)

FUNCTION CONTROL SELECT CONDITION

FUNCTION SELECT CONDITIONS

FUNCTION DRIVE

MATRIX ADDRESS SELECTION

Figure 17 (ISSUE 2)

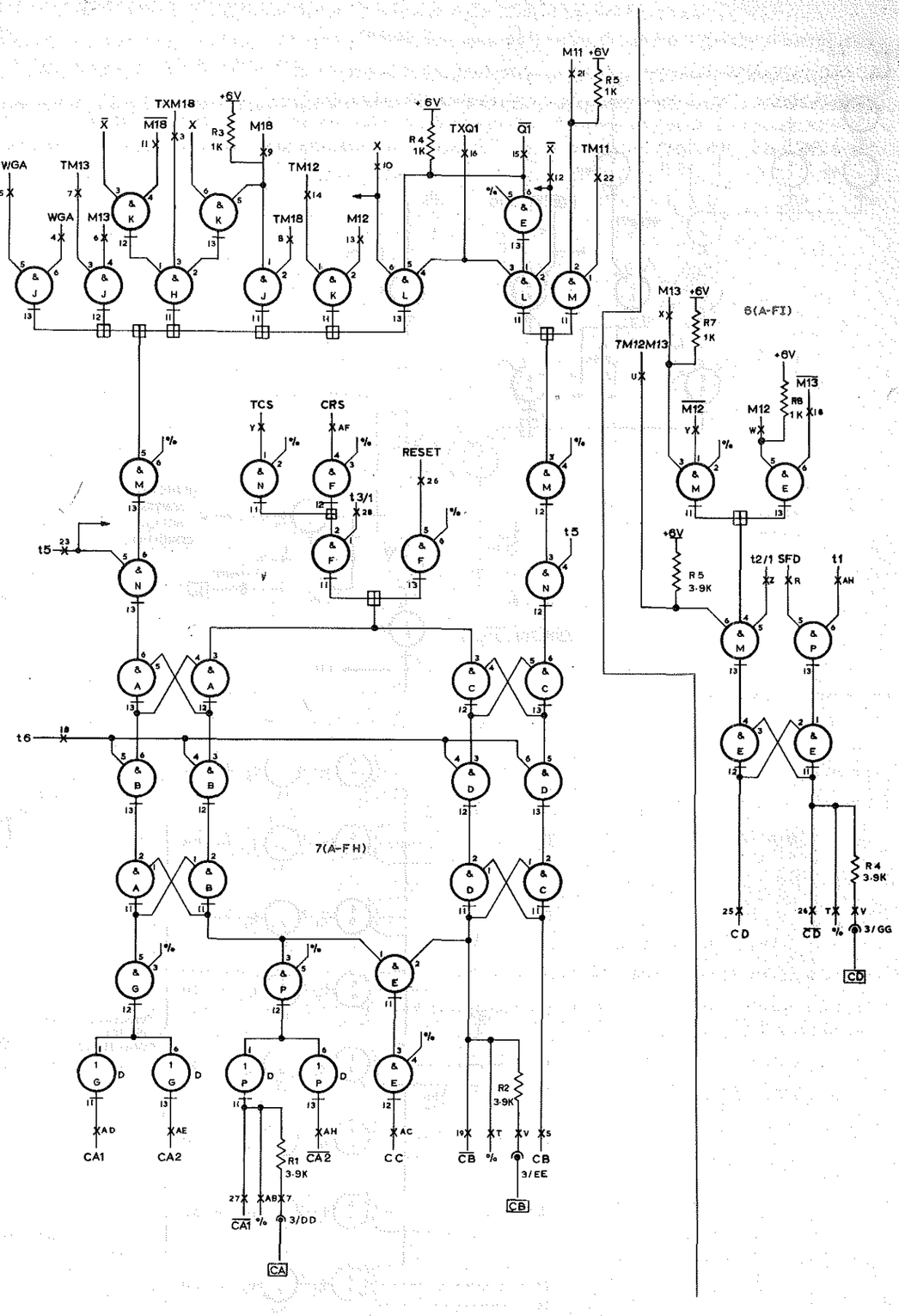
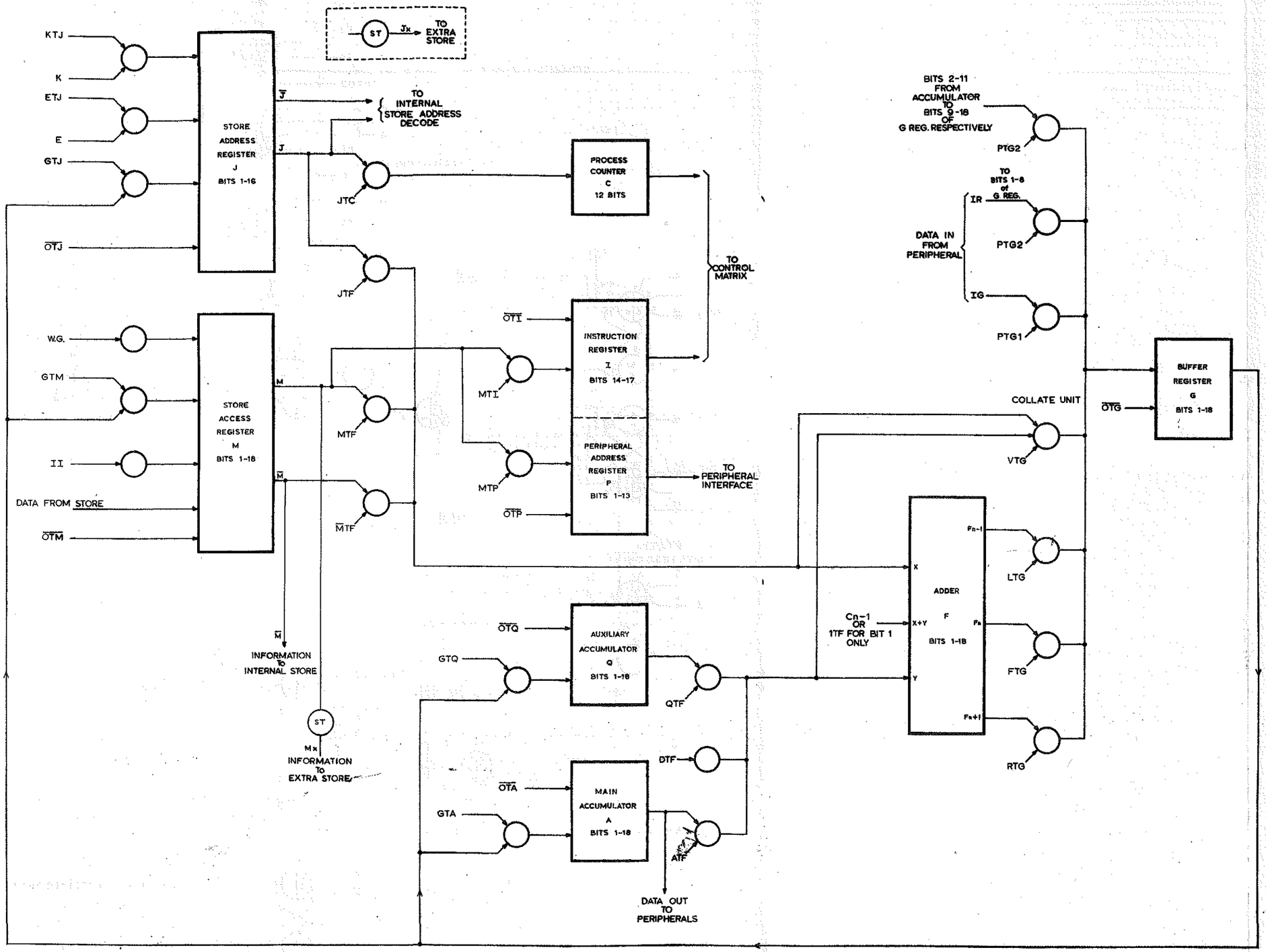


Figure 18 *(ISSUE 2)*

MATRIX CONDITIONS



TITLE	DESCRIPTION
$\overline{OTI}$, $\overline{OTJ}$ etc	All waveforms like these are register clear waveforms.
D T F	This waveform is divided into DTF/1 and DTF/2 and is used in conjunction with the collate unit. For more details see (FIG. 33 a, b & c)
I I	INITIAL INSTRUCTIONS bits from MATRIX AMPLIFIERS. See (FIG. 38 b).
W.G.	WORD GENERATOR
I.G.	This is an 18 bit data input from any peripheral device other than the paper tape reader.
I.R.	This is a 5,7 or 8 bit data input from the paper tape reader.
K	K is two waveforms K1 and K2 which are used to set up Process Counter for Mult. or Div. and K1 with E1 and E2 set up B-register address.
E	E is two waveforms E1 and E2 which are used to determine which SCR address or with K1 which Breg. address shall be selected according to the level of interrupt.

REGISTER SYSTEM

Figure 19 (ISSUE 2)

900
4.1.2.

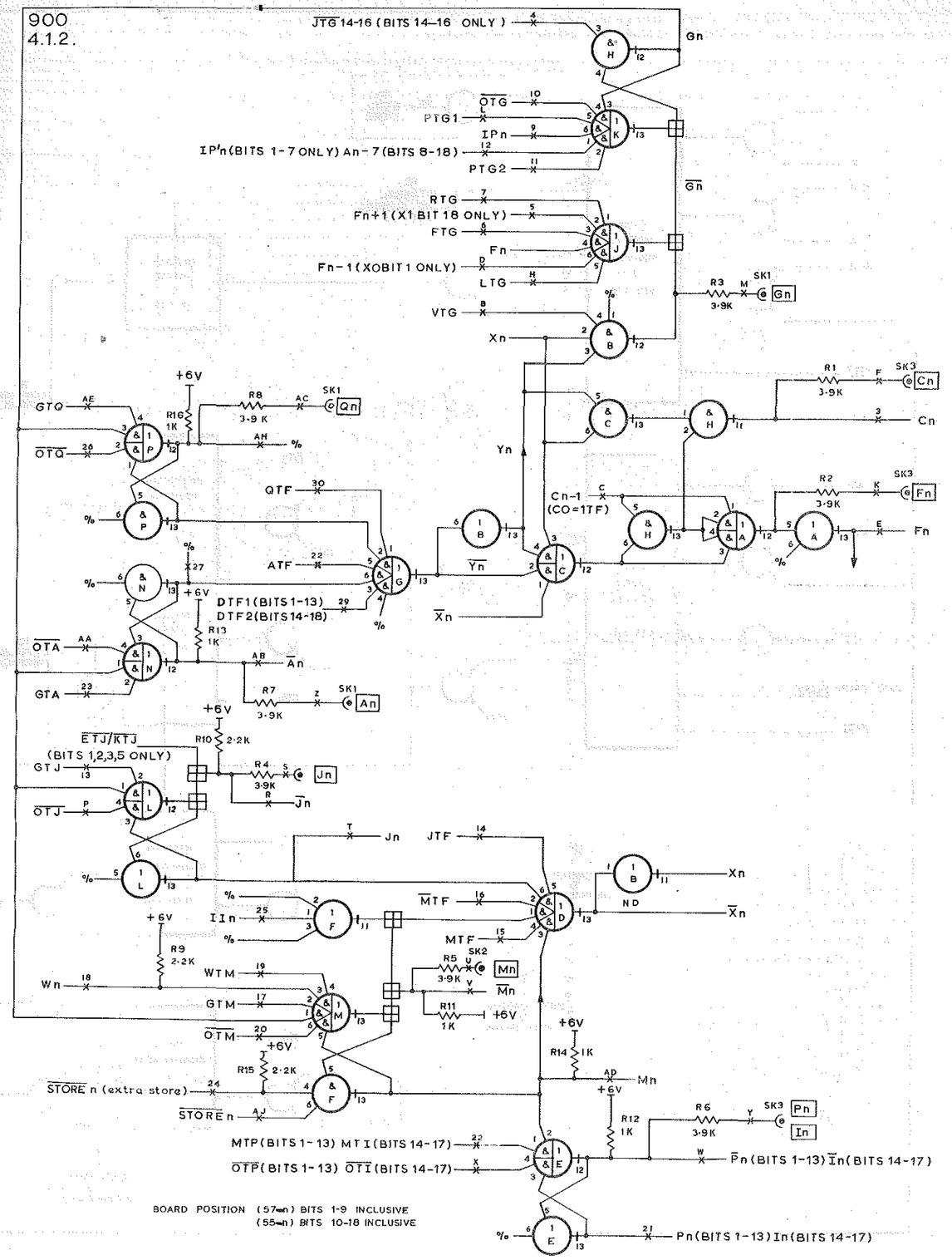
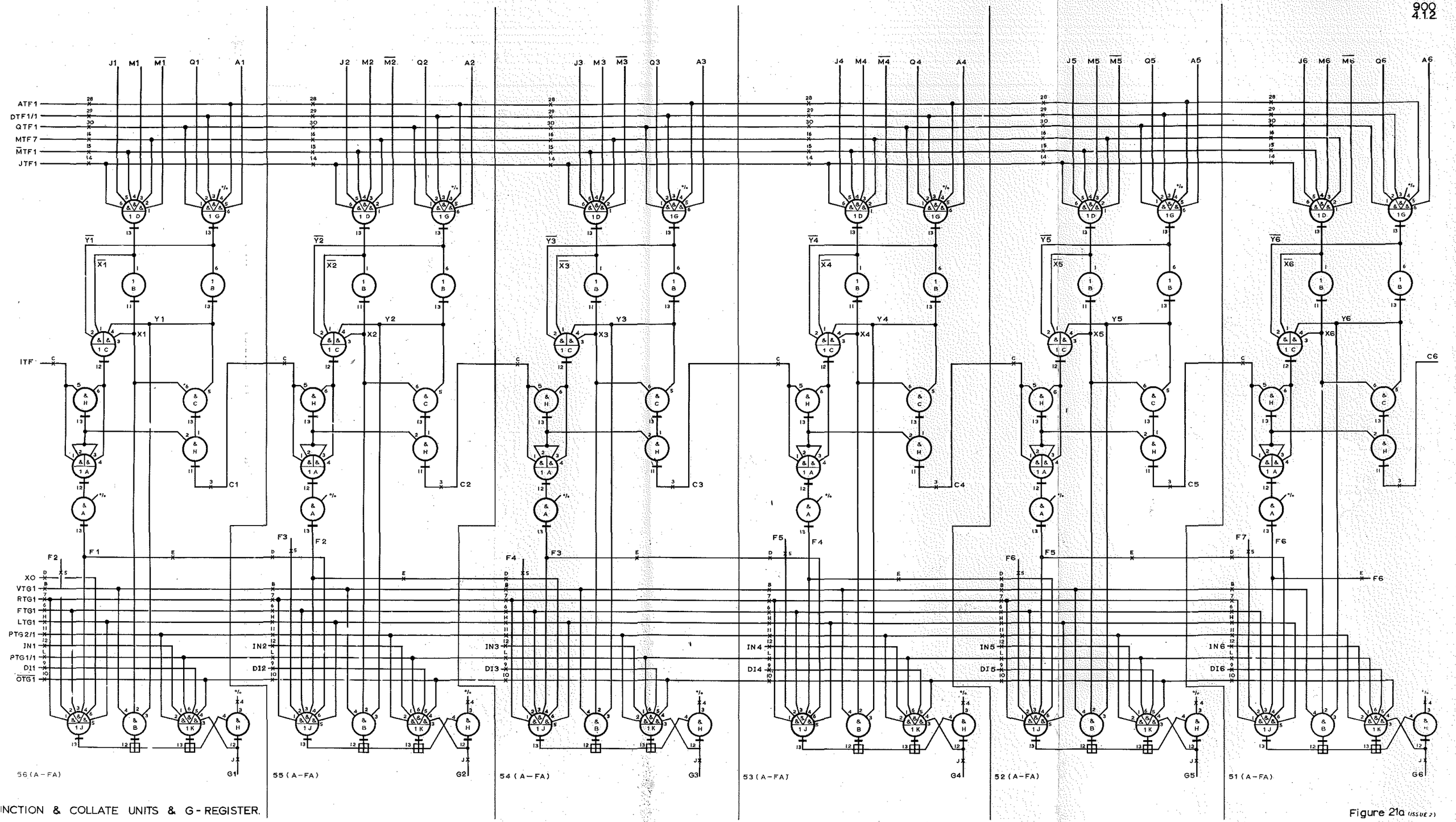


Figure 20 (ISSUE 2)

SAMPLE BIT OF REGISTERS AND FUNCTION UNIT



INCTION & COLLATE UNITS & G-REGISTER.

Figure 21a (ISSUE 2)

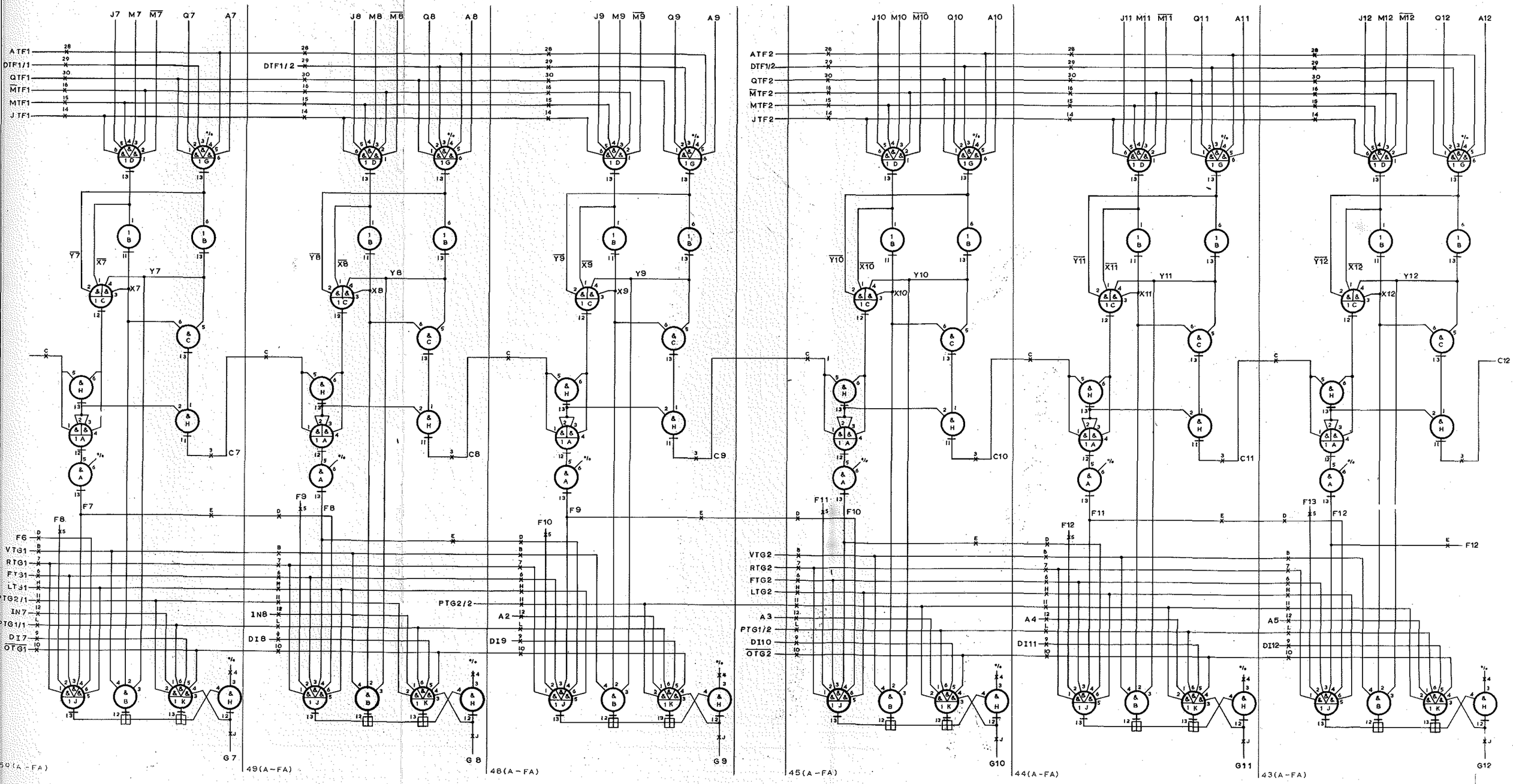
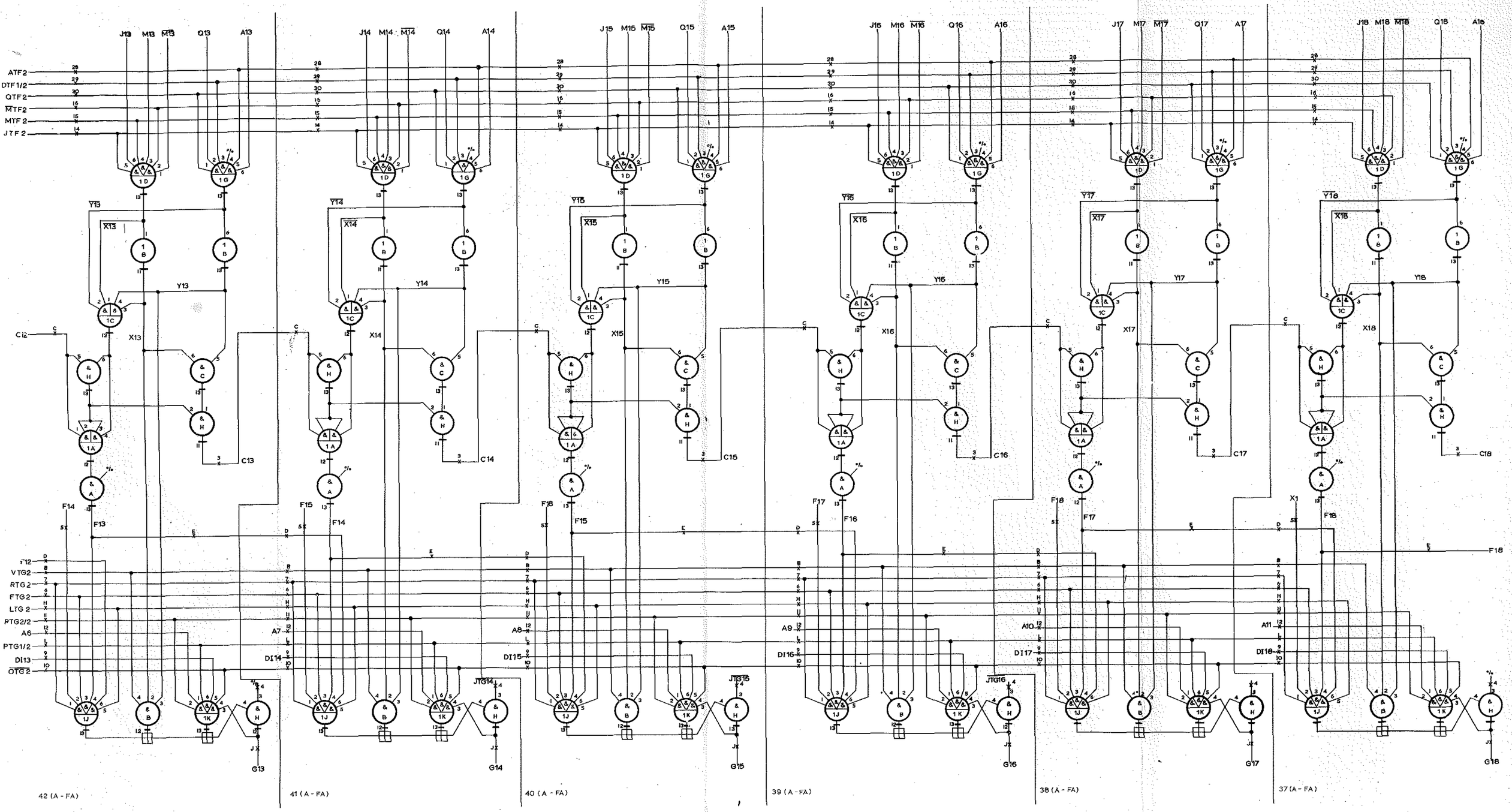


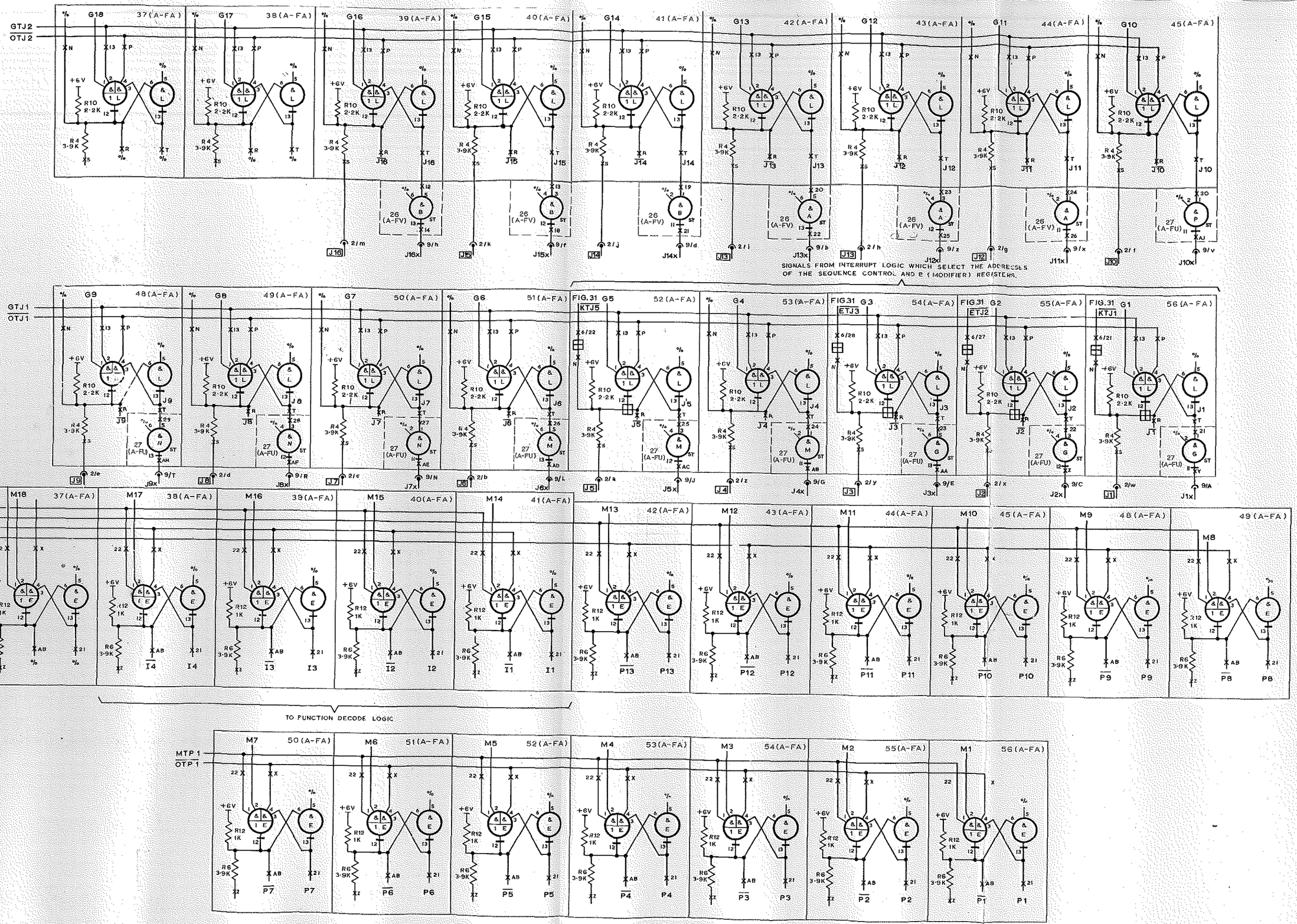
Figure 21b (ISSUE 2)

FUNCTION & COLLATE UNITS & G-REGISTER



FUNCTION & COLLATE UNITS. & G-REGISTER.

Figure 21c (issue 2)



& J REGISTERS.

Figure 22 (ISSUE 2)

47(A-FJ)

+6V
R4 3·9K
VTGa
t 2/2
AF
R1 3·9K
FTGa
t 2/2
R2 3·9K
LTGa
t 2/2
R3 3·9K
RTGa
t 2/2
R6 3·9K
PTG1a
t 2/2
R7 3·9K
PTG2a
t 2/2
&C &H &J &A &B &K
1C 1H 1J 1A 1B 1K
VTG 1
VTG 2
FTG 1
FTG 2
LTG 1
LTG 2
RTG 1
RTG 2
PTG1/1
PTG1/2
PTG2/1
PTG2/2

5(A-FR)

32(A-FN)
VTG
1K
S/P
AUDIO INNER

DTF1a
&E
1E
DTF1/2
DTF1/1

t'2
&M
1M
t2/2
t2/1
&D
1C
MTI
R8 3·9K
MTIa
R1 2·2K
t1'
&G
1G
t1
RESET
&H
R7 1K
OTI
t'3
&P
1P
t3/1
t3/2
R6 3·9K
JTCa
&A
46/Z
OTA'
TXQ1

22(A-FE)

R7 3·9K
R8 1K
ETJa
t4/2
&F
ETJ

OTA'
R10 3·9K
GTAa
t3/2
R4 2·2K
RESET/2
1M
OTA2
OTA1

46(A-FK)

R5 2·2K
t3/2
GTJa
&G

R3 3·9K
OTJa
t3/2
&K
OTJ'
RESET/2
&J
1J
OTJ2
OTJ1

R9 3·9K
R6 2·2K
OTMa
t3-2
RESET/2
1K
OTM2
OTM1

R3 2·2K
GTQa
t3-2
RESET/2
1P
OTQ1
OTQ2

R7 2·2K
MTPa
t1
RESET/2
1D
OTP2
OTP1

R14 3·9K
WTMa
t4/1
1C
WTM2
WTM1

t'4
1F
1G
t4/2

46(A-FK)

R13 3·9K
GTQa
t4/1
&N
1N
GTQ1
GTQ 2

R10 3·9K
GTAa
t4/1
1E
GTA 1
GTA 2

R11 3·9K
GTJ a
t4/1
1B
GTJ 1
GTJ 2

R12 3·9K
GTMa
t4/1
1L
GTM1
GTM 2

R8 2·2K
%
t1
RESET/2
&F
&A
1A
OTG 2
OTG 1

ATF a
&P
1P
ATF1
ATF2

JTF a
&D
1D
JTF1
JTF2

MTF a
&L
1L
MTF1
MTF2

QTF a
&N
1N
QTF1
QTF2

MTF a
&E
1E
MTF1
MTF2

R5 3·9K
MTP a
t2/2
&M
1M
MTP1
MTP2

47(A-FJ)

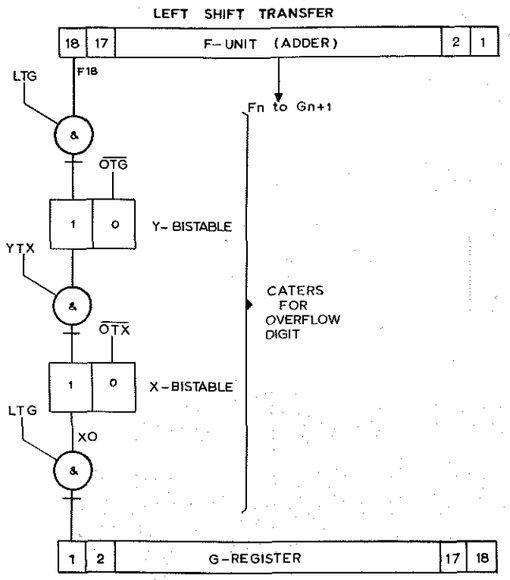
+6V
R8 3·9K
JTCa
t5
&F
1F
JTC1
JTC2

MTI
J14
J15
J16
&G
JTG14
JTG15
JTG16

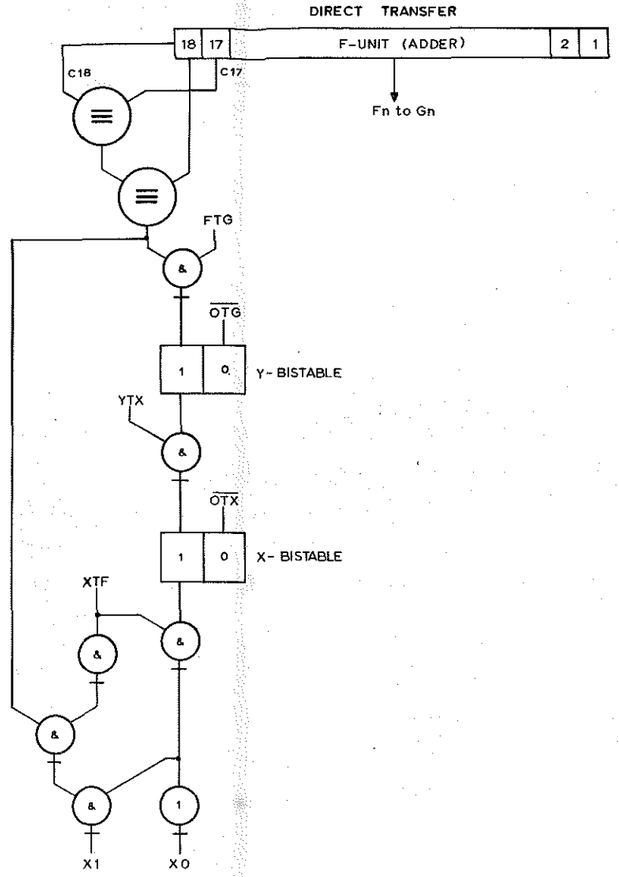
WEFORM DRIVERS

Figure 23 (ISSUE 2)

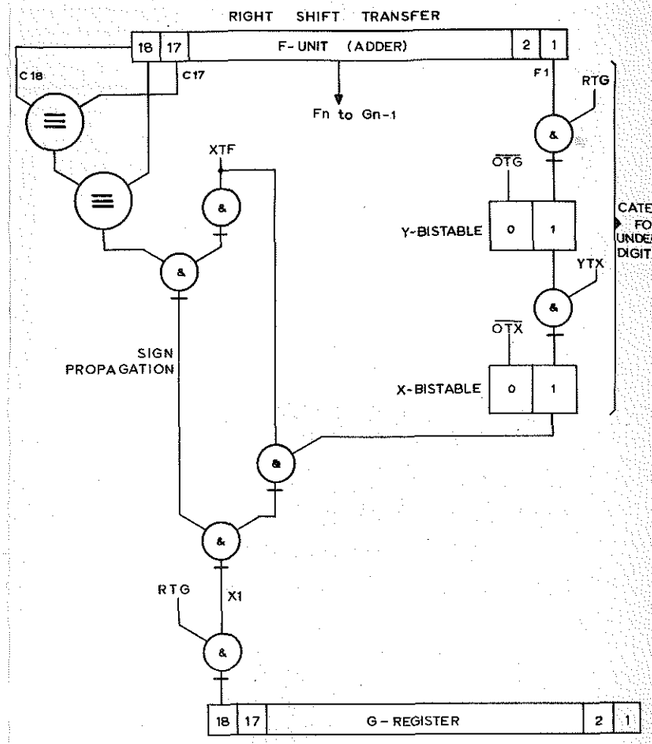
LEFT SHIFT TRANSFER



DIRECT TRANSFER



RIGHT SHIFT TRANSFER



5 (A-FR)

57 (A-FM)

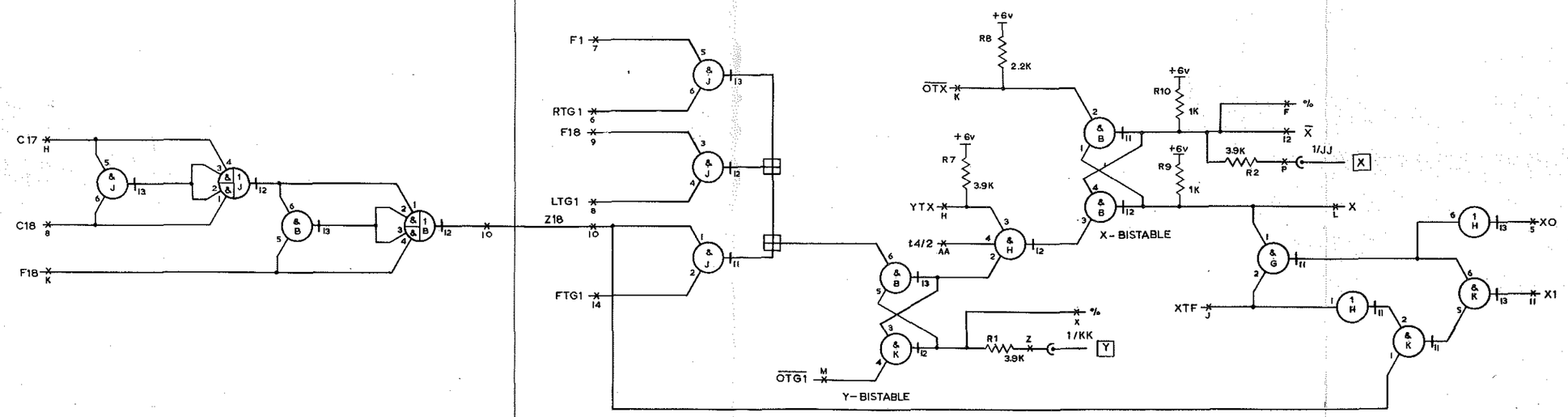


Figure 24 (ISSUE 2)

A-FE 22

1-SKT/

PC1 PC2 PC3 PC4

2/Z 2/AA 2/BB 2/CC

A-FP 23

PULSE GENERATOR MARGINS

+2·4v D

2·4v

PC1 PC2 PC3 PC4

R2 R1 R3 R4

3·9K

ITC

+2·4v

J 100 B 100 N 100 F 100

&J &K &G &P

&A &C &M &G

&2H &H &C &M &G

1 N

&N

1 N

C 1-4/1

1 L

&L

1 L

C1-4/0

$\overline{PC1}$ $\overline{PC2}$ $\overline{PC3}$ $\overline{PC4}$

JTC

$\overline{J1}$ $\overline{J2}$ $\overline{J3}$ $\overline{J4}$

SKT

PC5 PC6 PC7 PC8

2/DD 2/EE 2/FF 2/GG

A-FP 24

WFR

C5-12/0

$\overline{J5}$ $\overline{J6}$ $\overline{J7}$ $\overline{J8}$

SKT

PC9 PC10 PC11 PC12

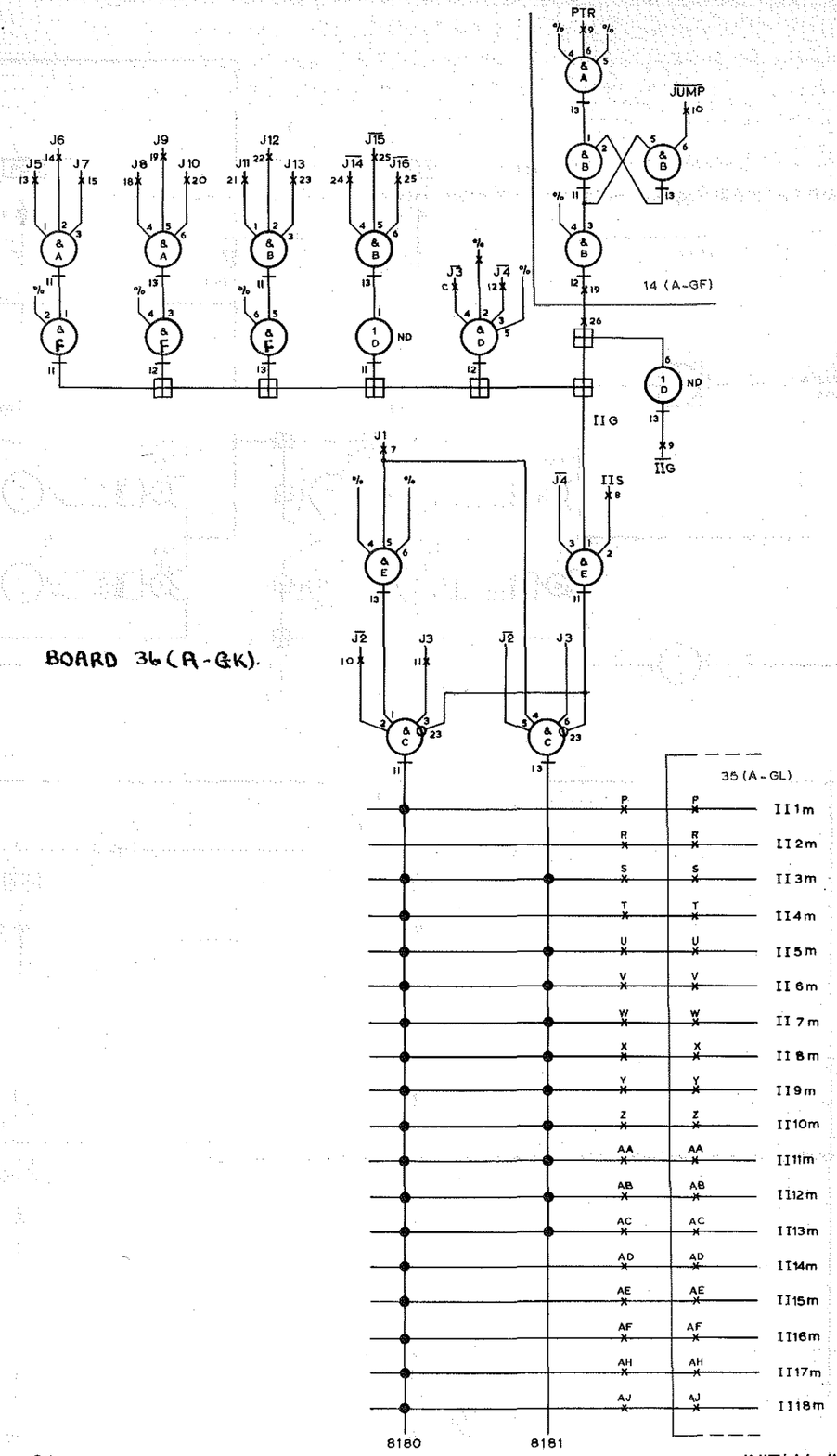
2/HH 2/JJ 2/KK 2/LL

A-FP 25

$\overline{J9}$ $\overline{J10}$ $\overline{J11}$ $\overline{J12}$

PROCESS COUNTER

Figure 25 (Issue 2)



BOARD 36(A-GK).

Figure 26a *(ISSUE 2)*

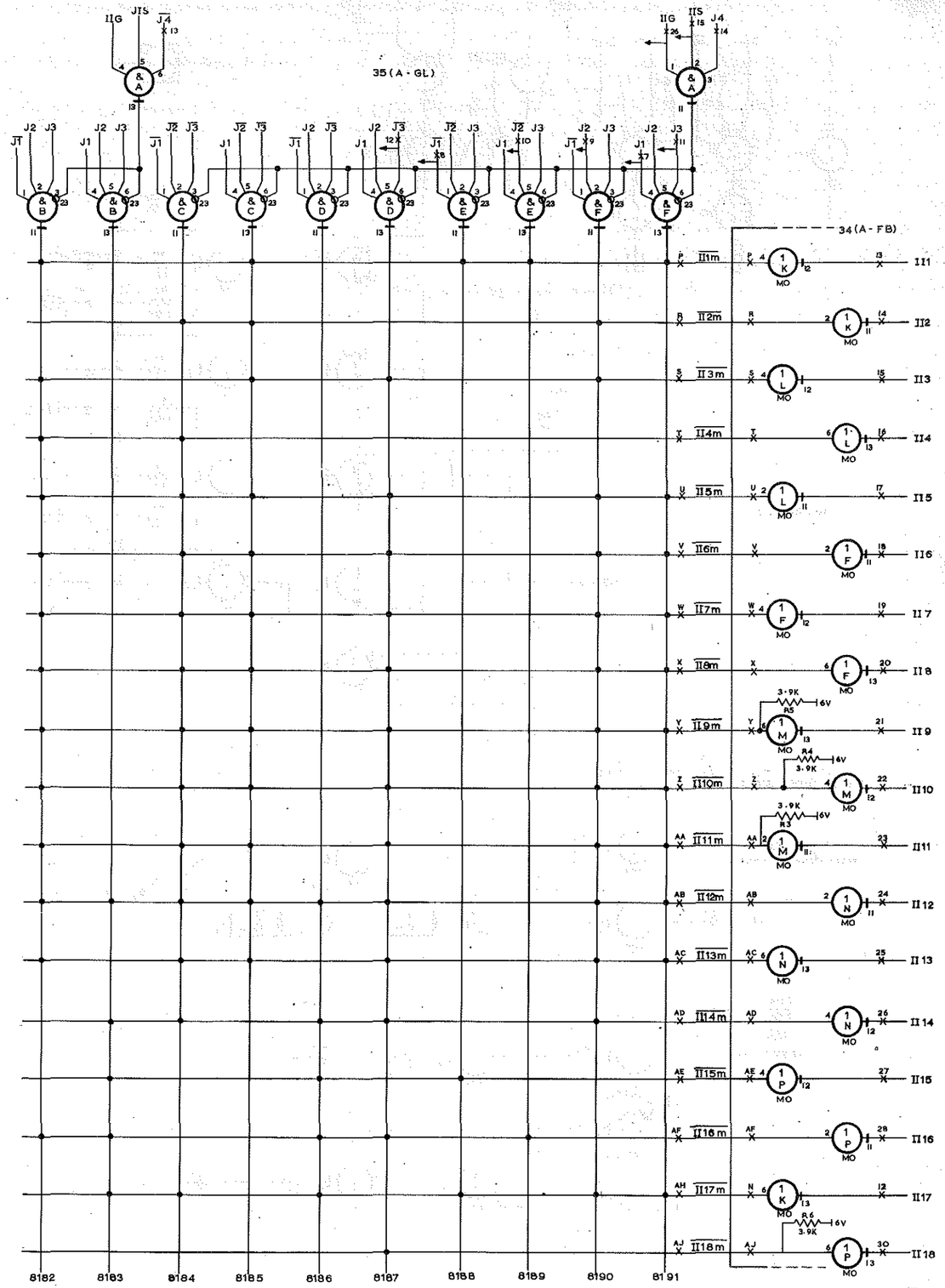


Figure 26b (ISSUE 2)

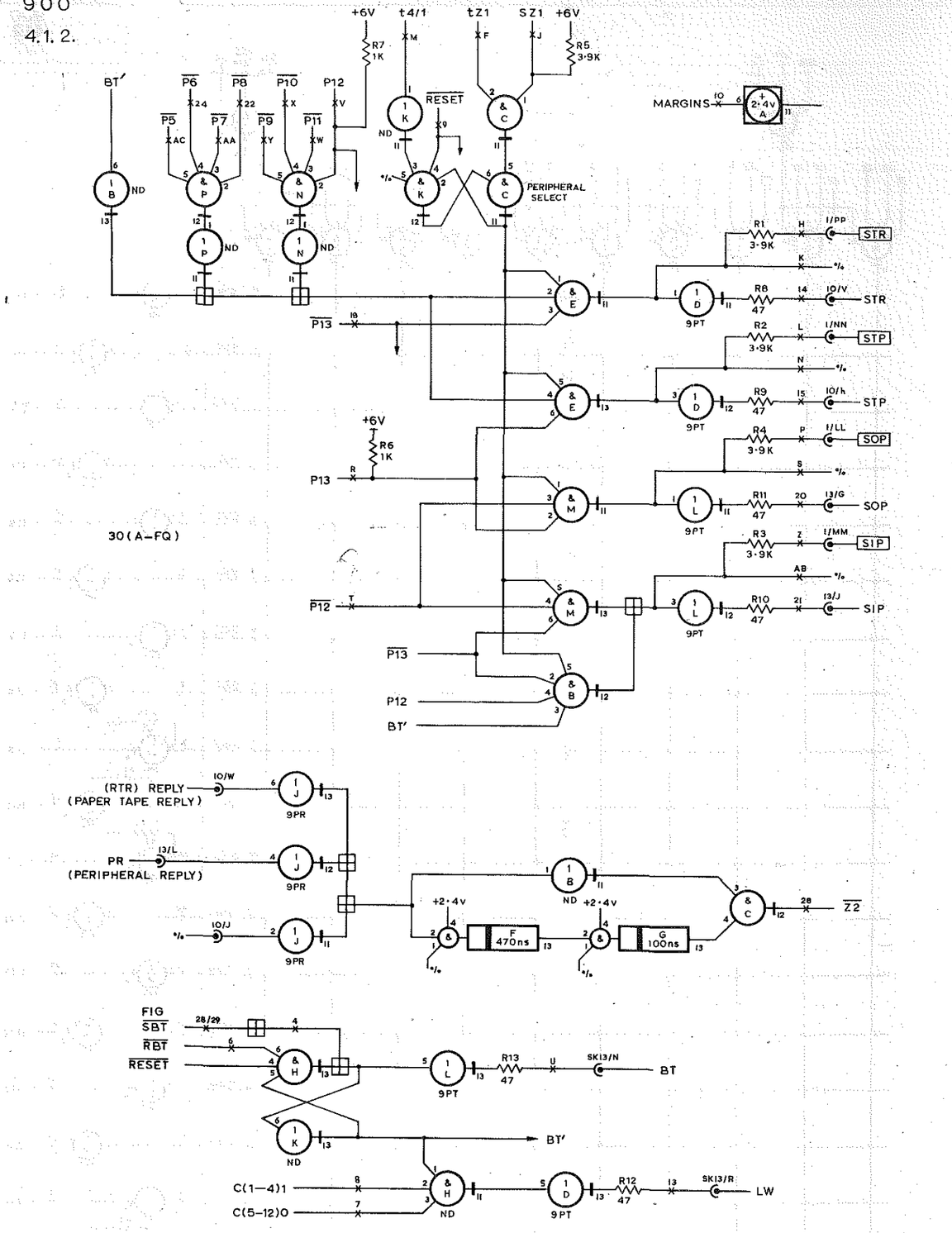
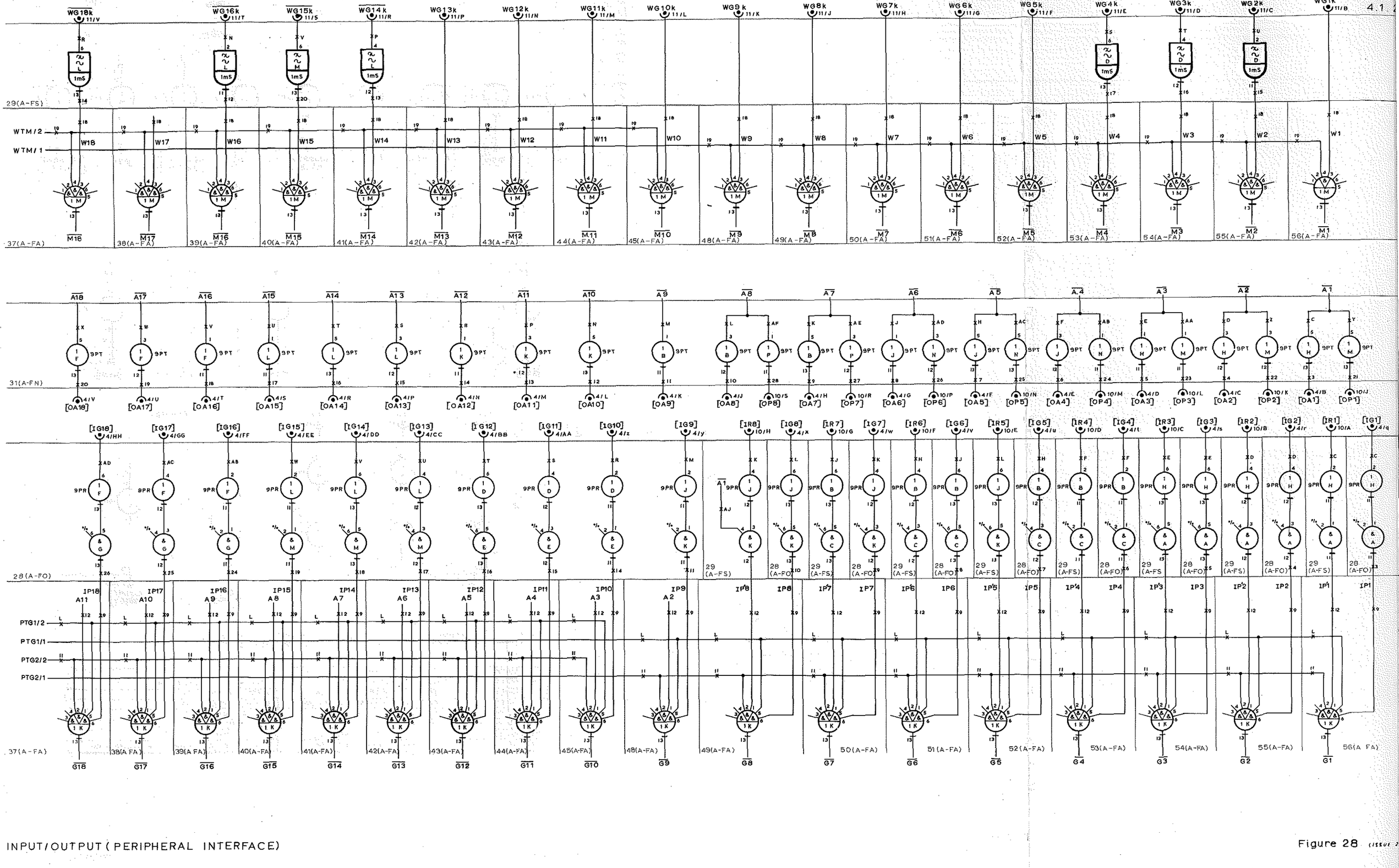


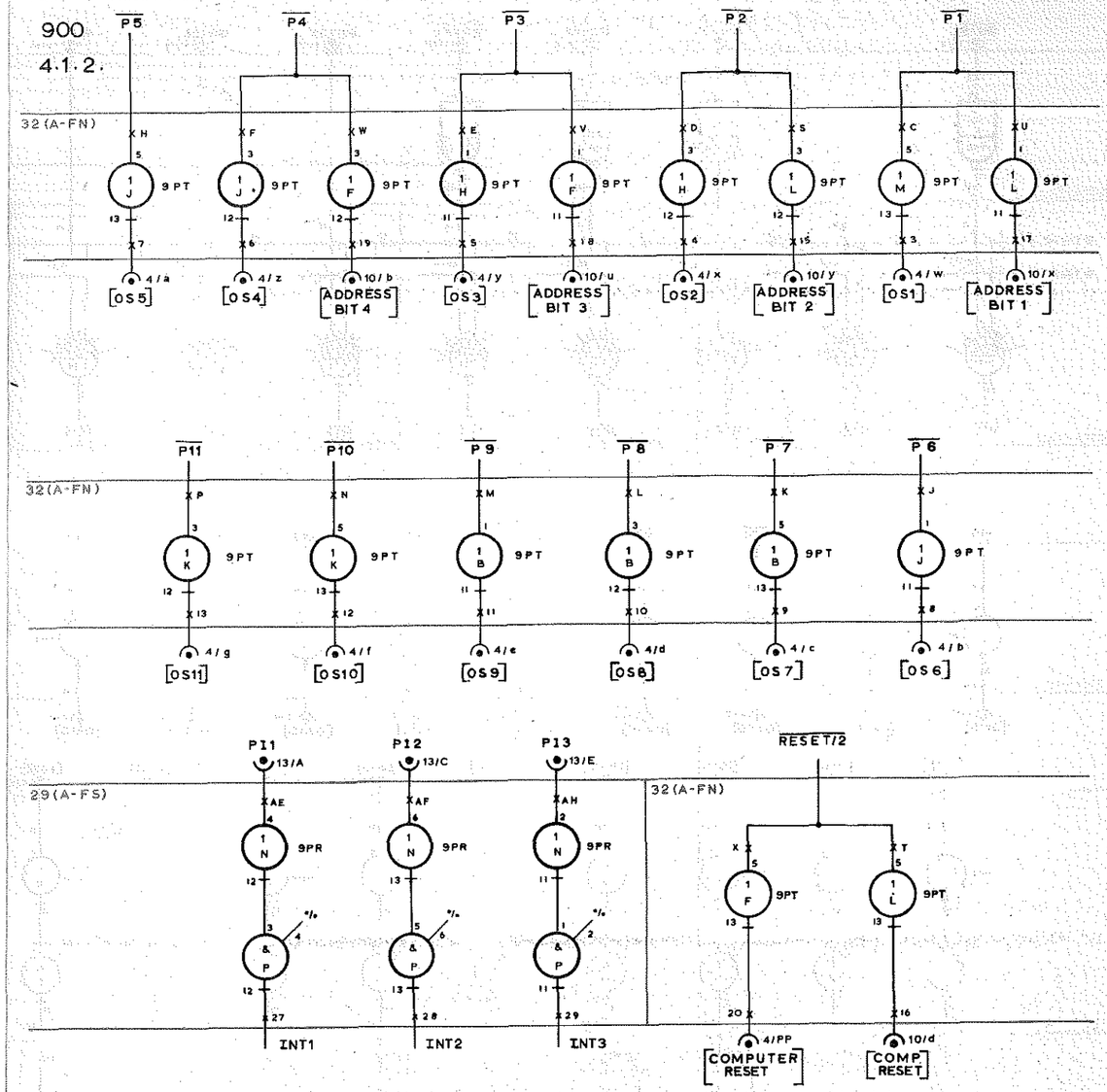
Figure 27 *(ISSUE 2)*



INPUT/OUTPUT (PERIPHERAL INTERFACE)

Figure 28 (issue 2)

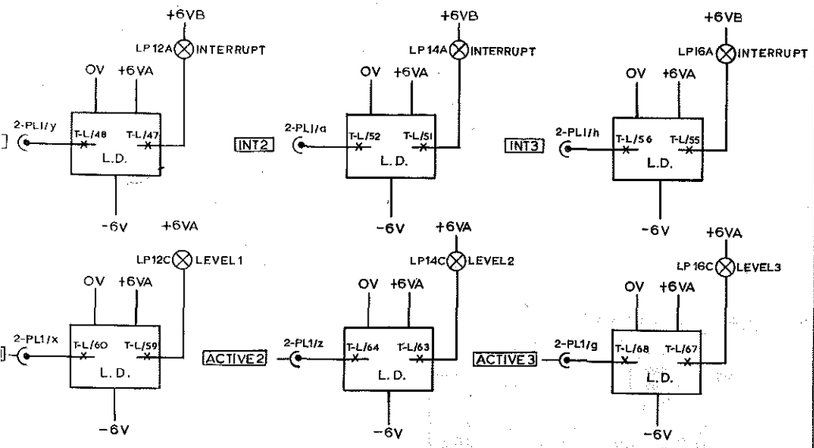
900
4.1.2.



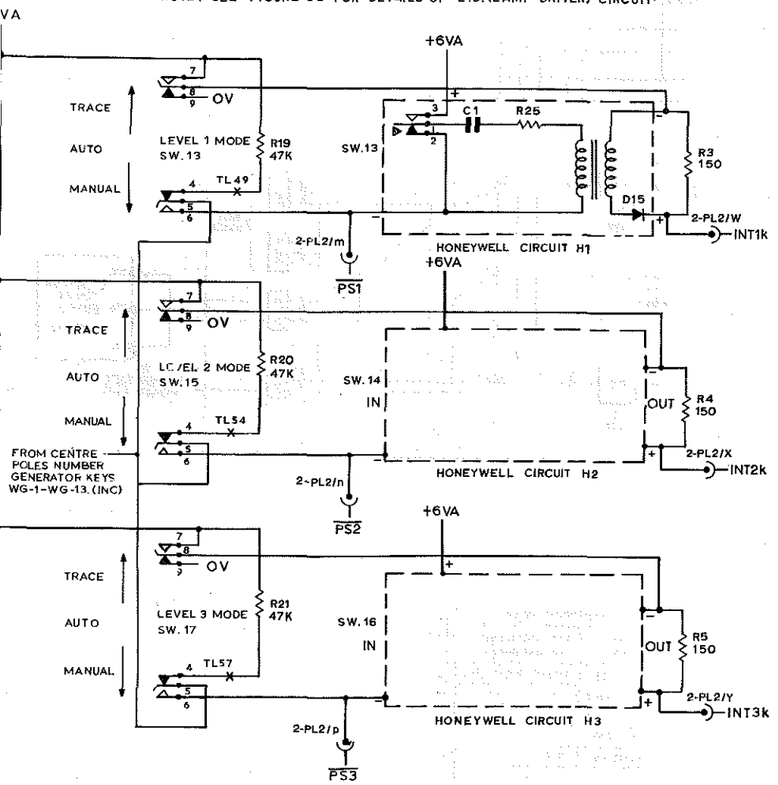
NOTE : ON FIGURE 29 SIGNALS WHICH ARE NOT RELEVANT TO INPUT/OUTPUT HAVE BEEN OMITTED FROM THE INPUT PINS OF THE PERTINENT ELEMENTS

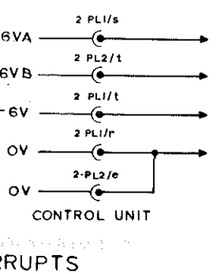
SIGNAL	FROM	TO
WG (INPUT)	WORD GENERATOR	STORE ACCESS REGISTER (M)
IG (INPUT)	GENERAL PERIPHERAL SOCKET (4)	BUFFER REGISTER (G)
IR (INPUT)	PAPER TAPE PERIPHERAL SOCKET (10)	BUFFER REGISTER (G)
OA (OUTPUT)	ACCUMULATOR	GENERAL PERIPHERAL
OP (OUTPUT)	ACCUMULATOR	PAPER TAPE PUNCH OR TELEPRINER
OS(INPUT & OUTPUT)	PERIPHERAL ADDRESS REGISTER	GENERAL PERIPHERAL
ADDRESS BIT	PERIPHERAL ADDRESS REGISTER	PAPER TAPE READER, PUNCH AND TELEPRINTER.

Figure 29 *(ISSUE 2)*



NOTE: SEE FIGURE 35 FOR DETAILS OF L.D.(LAMP DRIVER) CIRCUIT





PROGRAM LEVEL	INTERRUPT BISTABLES E1	E2	J REGISTER BITS SET J5	J3	J2	J1 SCR	J1 BREG
HIGHEST	1	0	0	0	0	0	1
	2	1	0	0	1	0	1
	3	0	1	0	1	0	1
BASE	4	1	1	0	1	1	0	1
MULTIPLY			1	0	0	1	
DIVIDE			1	0	1	0	

J-REGISTER SETTING SIGNAL	BIT AFFECTED
$\overline{KTJ1}$	J1
$\overline{ETJ2}$	J2
$\overline{ETJ3}$	J3
$\overline{KTJ5}$	J5

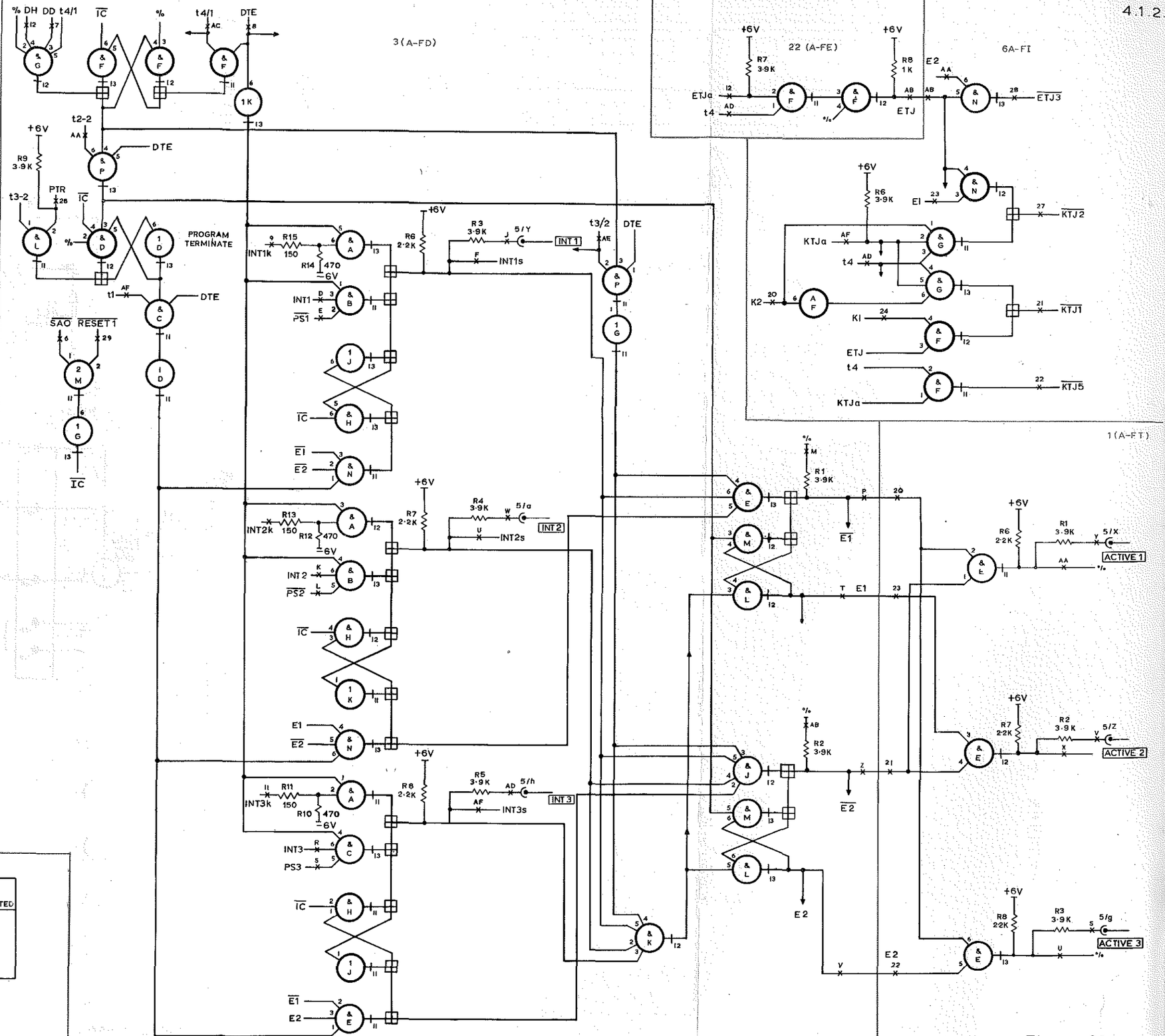
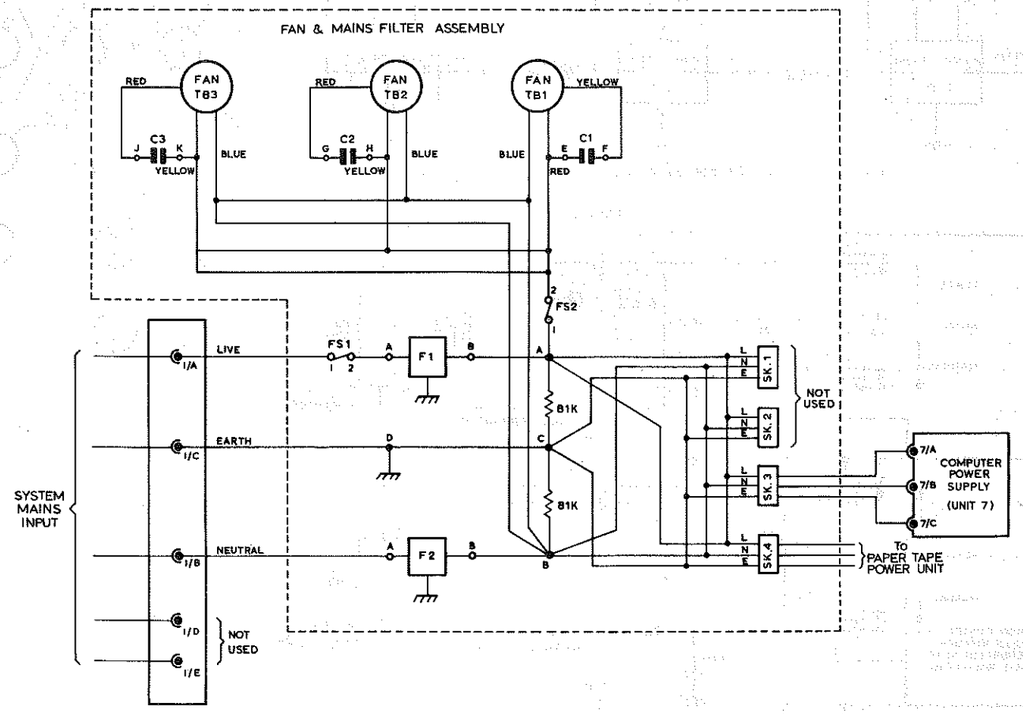
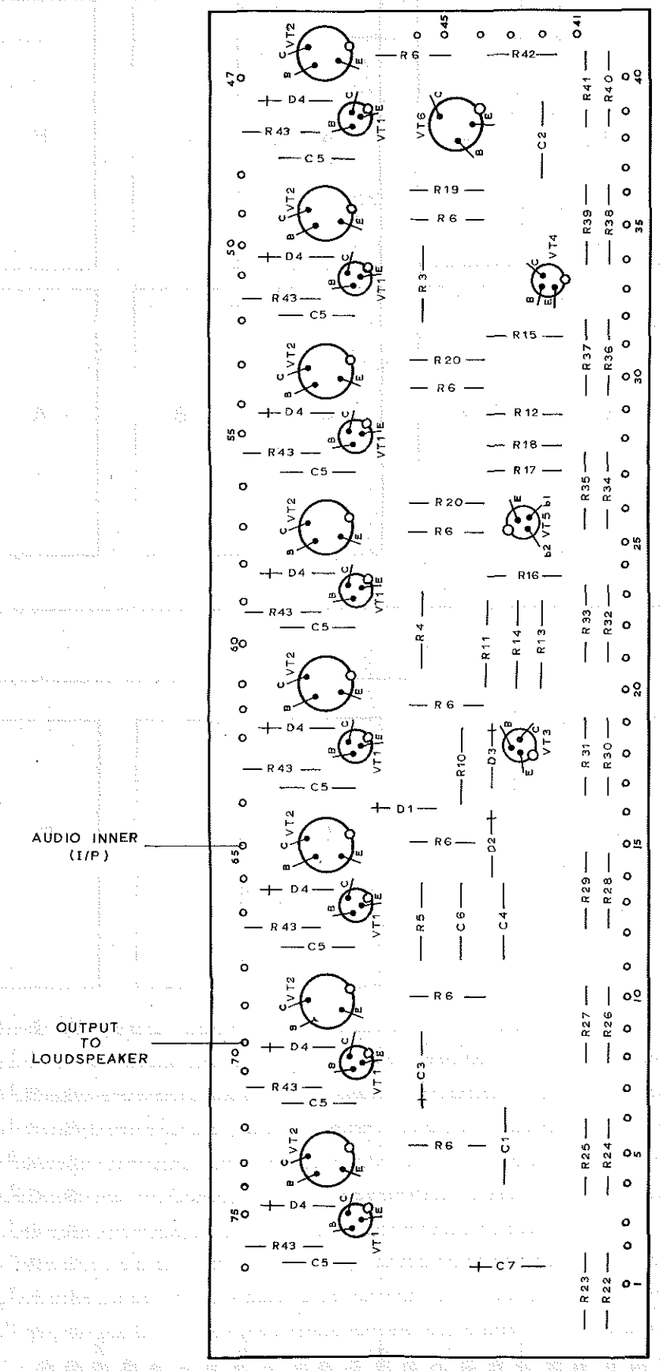


Figure 30 (ISSUE 2)



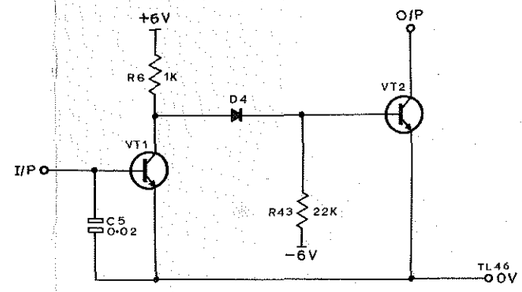
COMPONENT TABLE	
FS1 FS2	10A antisurge 3A antisurge
C1 – C3	1·0μF 1000V D.C.
Fan 1 - Fan 3	
FILTER 1&2	
SK1 – SK4	
Resistors	81K

Figure 31 *(ISSUE 2)*

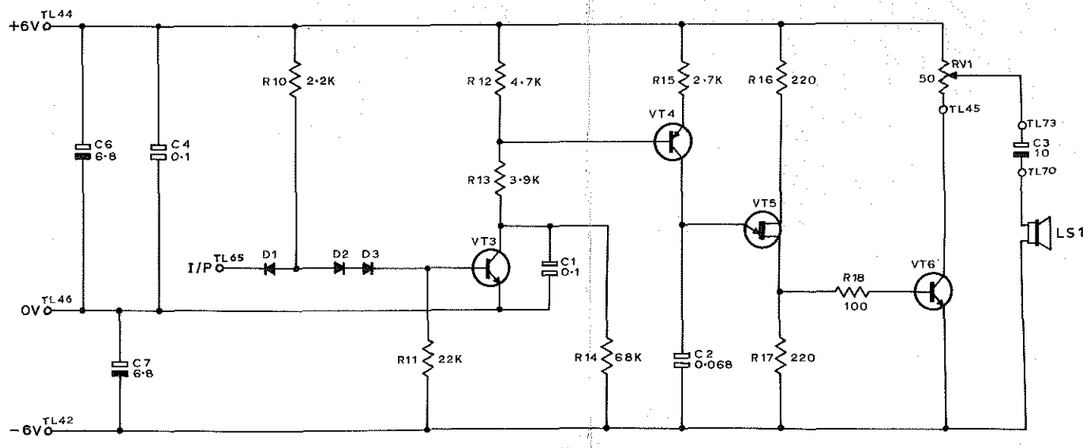


BOARD ASSEMBLY (CONTROL UNIT)

COMPONENT TABLE				
R.	VALUE	WATT. TOL%		
22–41	47	¼ ± 5	9219	
18	100	½ 5 TE	5662	
3–5	150	½ 5 TE	6205	
16–17	220	½ 5 TE	5885	
6	1K	½ 5 TE	5671	
10	2·2K	½ 5 TE	6017	
15	2·7K	½ 5 TE	5677	
13	3·9K	½ 5 TE	5679	
12	4·7K	½ 5 TE	5680	
11&43	22K	½ 5 TE	6012	
19–27	47K	½ 5 TE	5874	
14	68K	½ 5 TE	6389	
RV1	50Ω		6275	
D				
1–4			11606	
VT				
1,3			11832	
2,6			11607	
4			11608	
5			11609	



LAMP DRIVER CIRCUIT (8-OFF)



LOUDSPEAKER DRIVE CIRCUIT

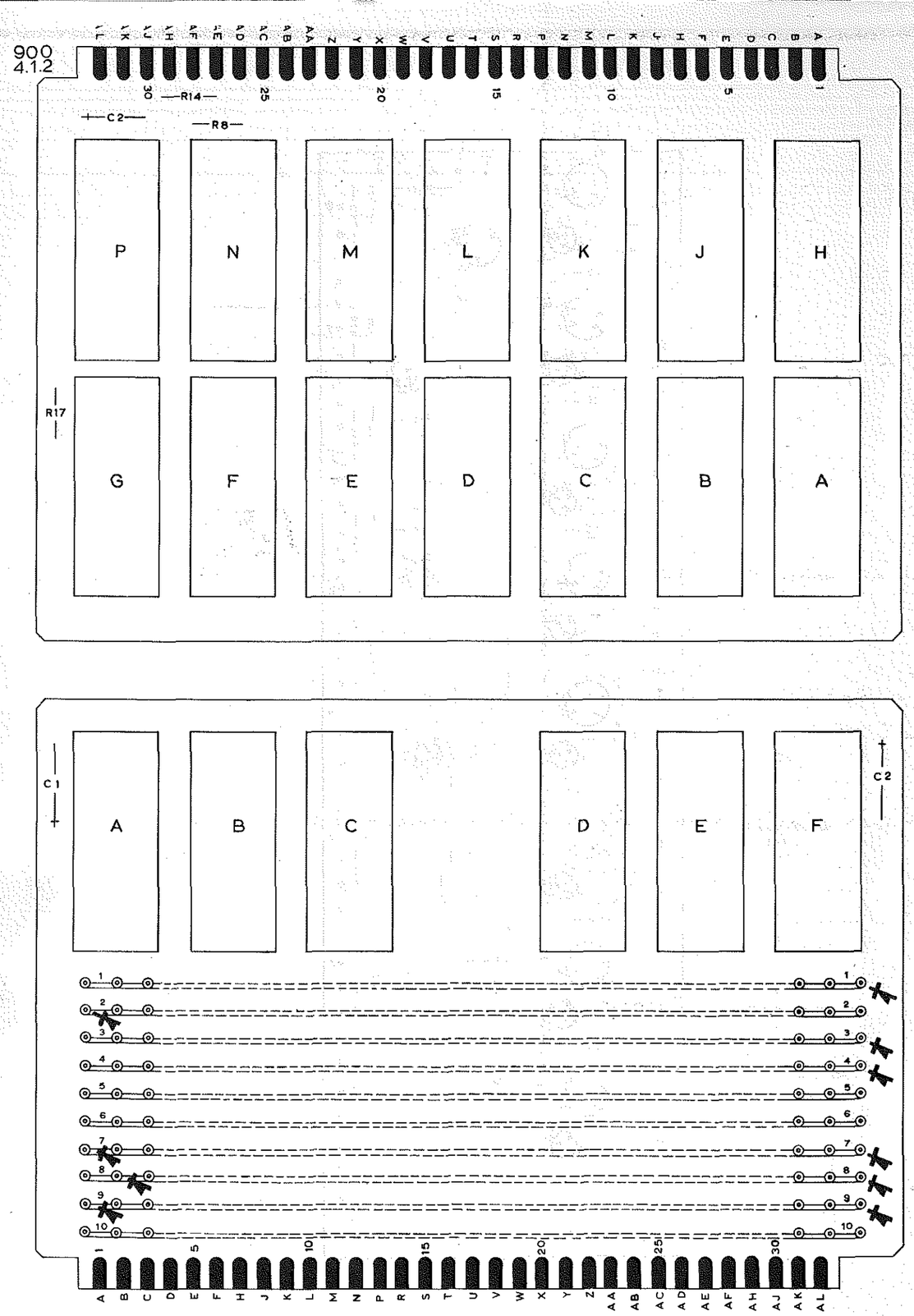


Figure 35 *(ISSUE 2)*

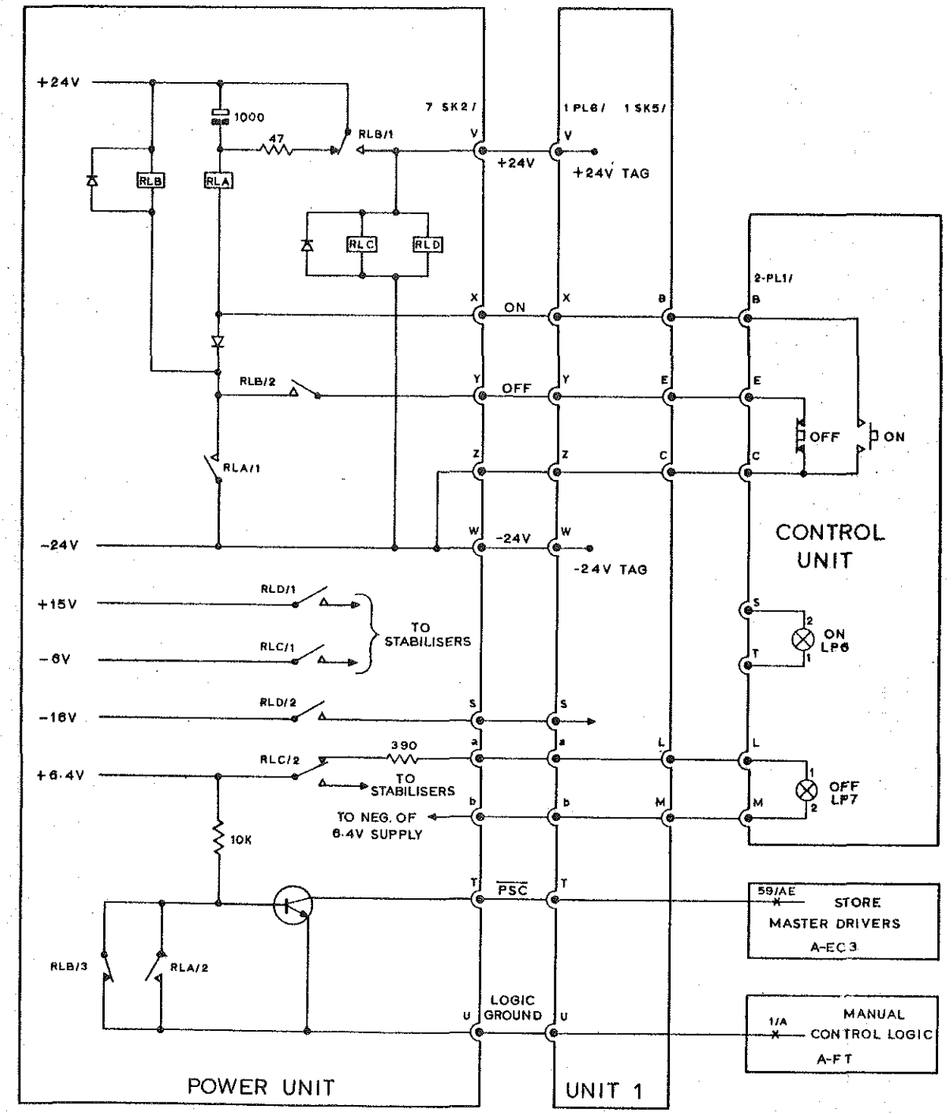
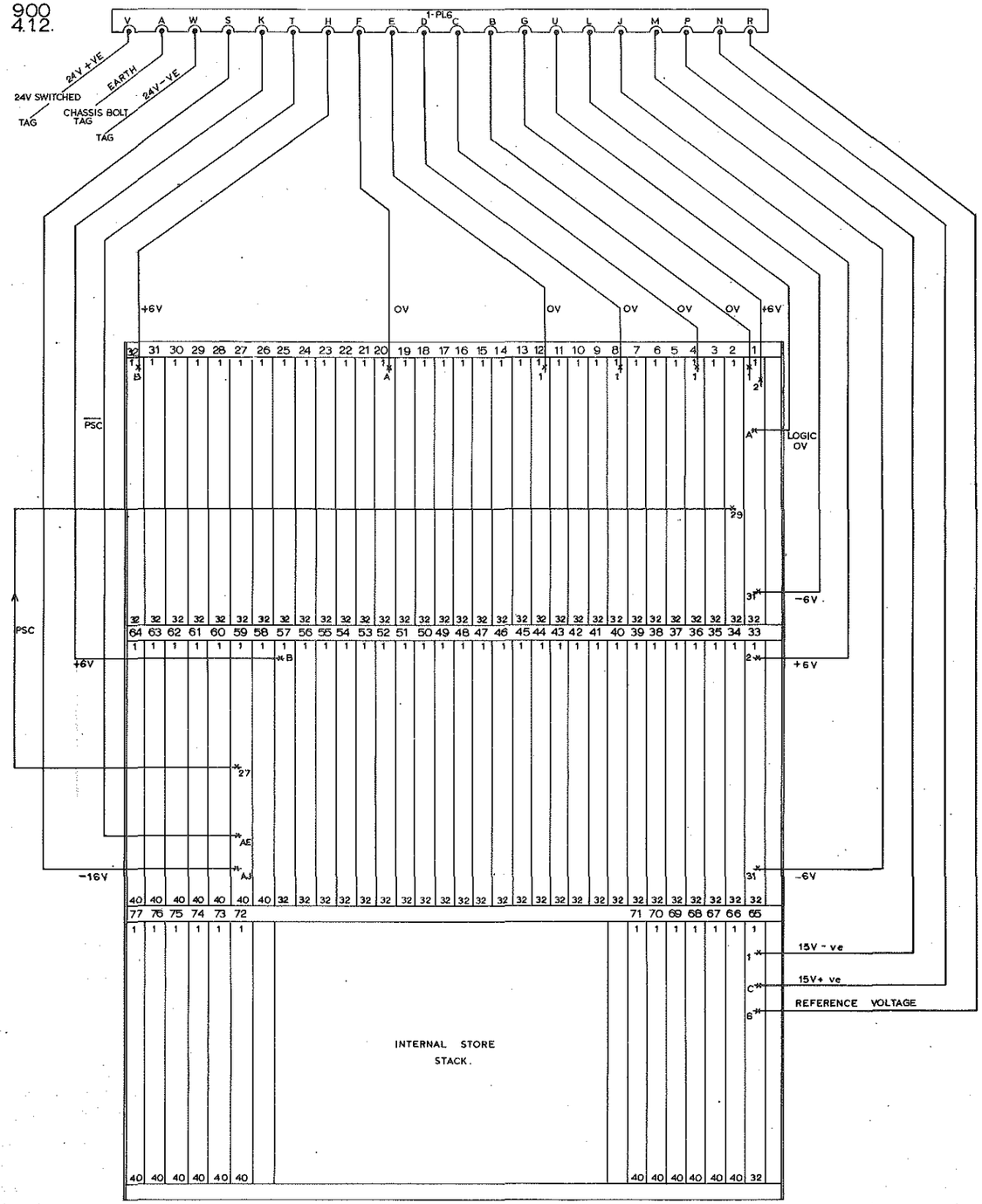


Figure 33 (ISSUE 2)



24V+ve	+6V	0V	TAG	-6V	-16V	24V-ve	15V+ve	15V-ve	
			A						SKT. 1 - DISPLAY UNIT PL1
		MM	A						SKT. 2 - DISPLAY UNIT PL 2
	MM	PP	A	NN					SKT. 3 - DISPLAY UNIT PL3
	P	LL	A						SKT. 4 - INPUT - OUTPUT UNIT
	s	r	A	t					SKT. 5 - CONTROL UNIT PL1
			A						SKT. 7 - MARGINAL TEST UNIT PL1
		g	A						SKT. 8 - EXTRA STORE
g		j	A			h			SKT. 9 - EXTRA STORE
f		e	T			g			SKT. 10 - PAPER TAPE CONTROLLER PL2
	t		A						SKT. 11 - CONTROL UNIT PL2
HH			JJ			KK			SKT. 14

Figure 32 (ISSUE 2)

D.C. DISTRIBUTION

L.S.A. AREA	L.S.A. BOARD TYPES																																
	A FA	A FB	A FC	A FD	A FE	A FF	A FG	A FH	A FI	A FJ	A FK	A FL	A FM	A FN	A FO	A FP	A FQ	A FR	A FS	A FT	A FU	A FV	A GA	A GB	A GC	A GD	A GE	A GF	A GG	A GH	A GI	A GJ	A GK	A GL
A	06		08	01	03	01	15	01	02	15	15	08	01		01	01	08	03	01	01	22	22	02		01	01	02	02	02	02	02	02	03	02
B	03	04	01	02	02	02	15	01	01	15	15	01	01	11	12	07	03	06	12	01	22	22	02		02	01	02	01	02	02	02		02	02
C	06		07	02	02	01	15	01	03	15	15	13	08		01	01	01	15	01	14	22	22	01		01	01	02	02	02	02	02	02	02	02
D	05		09	03	01	01	01	01	18	15	15	03	13		12	08	11	01	18	03	22	22			01	01	02	02	01	02	02	02	02	02
E	06		02	02	01	02	01	01	01	15	15	03	03		01	01	02	15	23	01	22	23	02		01	01	01	02	02	02	02	02	02	02
F	02	04	01	01	01	01	15	01	01	15	01	01	03	11	12	07	09	15	12	18	22	23	01		01			01	02	02	02	02	01	02
G	05		01	03	03	01	15	15	02	01	01	01	01		01	01	13	15	01	01	22	23												
H	01	04	03	01	03	02	15	02		15	04	01	03	11	12	01	02	01	12	18		23												
J	05	04	13	03	03	01	15	01	03	15	15	13	01	11	12	07	12	06	12	18		23												
K	05	04	01	03	03	01	15	01	15	15	15	03	01	11	01	01	03	01	01	18		23												
L	06	04	01	01	02	02	15	02	01	15	15	01	01	11	12	03	11	03	18	01														
M	05	04	07	01	02	01	15	01	02	15	15	03	01	11	01	01	02	15	18	18	22													
N	06	04	07	02	03	02	15	01	01	15	15	01	02	11	04	07	03	15	12	01	22													
P	06	04	07	02	01		15	15	01	15	15	01	02	11	01	01	03	15	01	01	22													
																	R8, R13																	
				R11, R13 R15																														
			R2 – R7	R10, R12 R14																														
	R10–R14 R16				R2 – R5 R8			R3 – R5	R7, R8				R9, R10				R6, R7	R7															
	R9, R15		R8	R6 – R8	R9	R7 – R9					R3 – R8	R7–R10	R3, R8 R11–R13					R1, R2		R4–R10													
	R3 – R8	R1 – R6	R1	R1 – R5 R9	R6, R7	R1 – R6		R1, R2	R1 – R6	R1 – R8	R9–R14	R1 – R5	R1, R2 R4 – R7			R1 – R4	R1 – R5	R3 – R6 R8		R1 – R3													
												R6																					
												C3																					
	C1 C2	C1 C2	C1 C2	C1 C2	C1 C2	C1 C2	C1 C2	C1 C2	C1 C2	C1 C2	C1 C2	C1 C2	C1 C2	C1 C2	C1 C2	C1 C2	C1 C2	C1 C2	C1 C2	C1 C2	C1 C2	C1 C2	C1 C2	C1 C2	C1 C2	C1 C2	C1 C2	C1 C2	C1 C2	C1 C2	C1 C2	C1 C2	C1 C2	C1 C2

DISCRETE COMPONENTS			
COMPONENT	VALUE	TOL ± %	CAT. No.
RESISTOR	47	5	9219
"	150	5	9197
"	470	5	9169
"	1K	5	9159
"	2·2K	5	9160
"	3·9K	5	11229
"	100K	5	11363
CAPACITOR	4·7		11750
"	6·8		11751